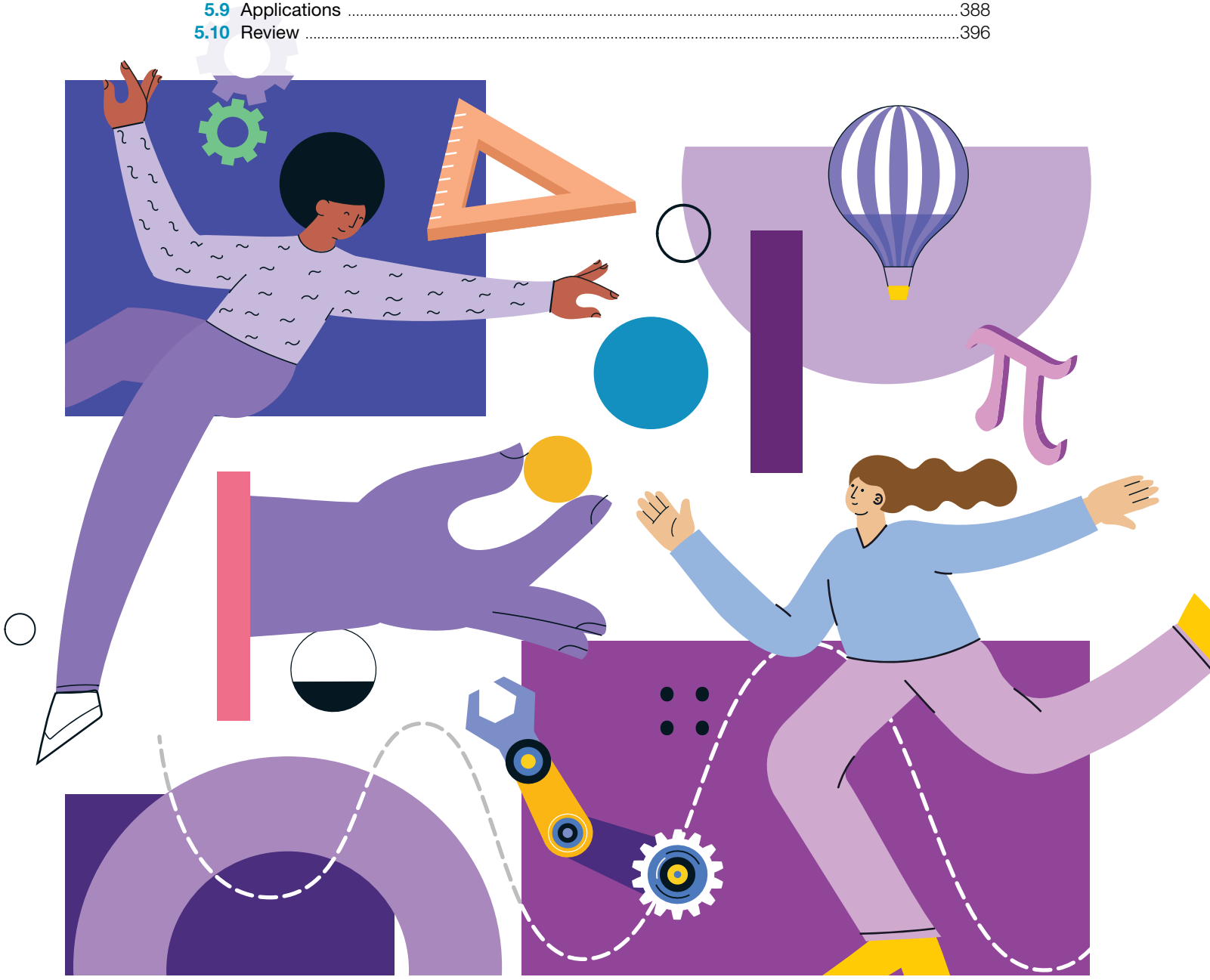
5 Trigonometry I

LESSON SEQUENCE

5.1 Overview 338
5.2 Pythagoras' theorem 341
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5.4 Trigonometric ratios 356
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5.6 Using trigonometry to calculate angle size 369
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LESSON
5.1 Overview

Why learn this?



Nearly 2000 years ago Ptolemy of Alexandria published the first book of trigonometric tables, which he used to chart the heavens and plot the courses of the Moon, stars and planets. He also created geographical charts and provided instructions on how to create maps.

The word trigonometry is derived from the Greek words 'trigonon' and 'metron', meaning *triangles* and *measure* respectively. This field of mathematics has reportedly been studied across the world since the third century BCE, with major discoveries made in India, China, Greece and Persia. The works ranged from developing relationships, axioms and proofs to describing its application to everyday use and life.

Trigonometry is the branch of mathematics that makes the whole universe more easily understood. The role and use of trigonometry in navigation in the early years were crucial and its application and study grew from there. Today, it is used in architecture, surveying, astronomy and, as previously mentioned, navigation. It also provides the foundation of the study of sound and light waves, resulting in its application in the areas of music manufacturing and composition, study of tides, radiology and many other fields.

Hey students! Bring these pages to life online



Watch videos

Engage with interactivities

Answer questions and check solutions

Find all this and MORE in jacPLUS

Reading content and rich media, including interactivities and videos for every concept



Extra learning resources

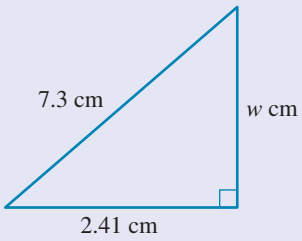
Differentiated question sets

Questions with immediate feedback, and fully worked solutions to help students get unstuck

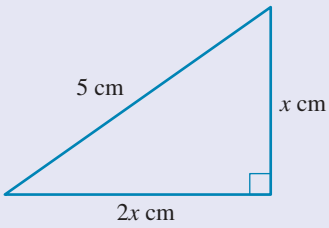
Exercise 5.1 Pre-test

learnon

1. Determine the value of the pronumeral w, correct to 2 decimal places.

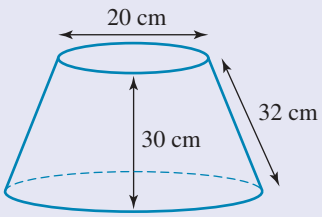


2. Determine the value of x, correct to 2 decimal places.



3. A square-based pyramid is 16 cm high. Each sloping edge is 20 cm long. Calculate the length of the sides of the base, in cm correct to 2 decimal places.

4. A cork is in the shape of a truncated cone; both the top and the base of the cork are circular.

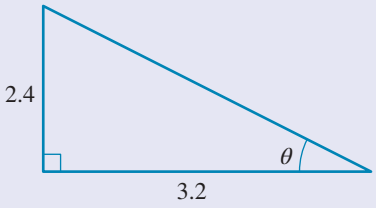


Calculate the sum of diameters of the top and the base. Give your answer in cm to 2 decimal places.

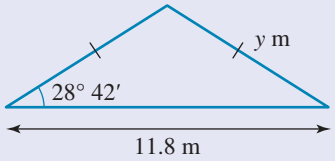
5. Evaluate $\sin(20°37')$, correct to 4 decimal places.

6. Calculate the size of the angle θ, correct to the nearest minute, given that $\cos(\theta) = 0.5712$. Give your answer in degrees and minutes.

7. Determine the size of the angle θ, correct to the nearest second.



8. Calculate y, correct to 1 decimal place.



9. MC Tyler is standing 12 m from a flagpole and measures the angle of elevation from his eye level to the top of the pole as 62°.
The distance from Tyler's eyes to the ground is 185 cm. The height of the flagpole correct to 2 decimal places is:

A. 20.71 metres **B.** 22.56 metres
C. 24.42 metres **D.** 207.56 metres

10. Change each of the following compass bearings to true bearings.

a. N20°E **b.** S47°W **c.** N33°W **d.** S17°E

11. MC In a right square-based pyramid, the square base has a length of 7.2 cm. If the angle between the triangular face and the base is 55°, the angle the sloping edge makes with the base is:

A. 45.3° **B.** 55°
C. 63.4° **D.** 47.7°

12. MC A boat travels 15 km from A to B on a bearing of 032°T. The bearing from B to A is:

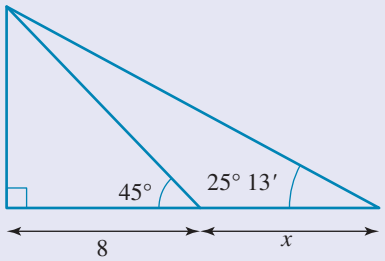
A. 032°T **B.** 058°T
C. 122°T **D.** 212°T

13. A bushwalker travels N50°W for 300 m and then changes direction 220°T for 0.5 km. Determine how many metres west the bushwalker is from his starting point.
Give your answer in km correct to 1 decimal place.

14. MC P and Q are two points on a horizontal line that are 120 metres apart. If the angles of elevation from P and Q to the top of the mountain are 34°5′ and 41°16′ respectively, the height of the mountain correct to 1 decimal place is:

A. 81.2 metres **B.** 105.3 metres
C. 120.5 metres **D.** 354.7 metres

15. Determine the value of x in the following figure, correct to 1 decimal place.



LESSON
5.2 Pythagoras' theorem

LEARNING INTENTIONS

At the end of this lesson you should be able to:

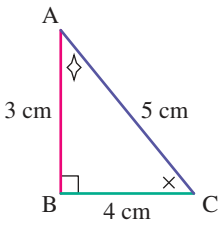
- identify similar right-angled triangles when corresponding sides are in the same ratio and corresponding angles are congruent
- apply Pythagoras' theorem to calculate the third side of a right-angled triangle when two other sides are known.

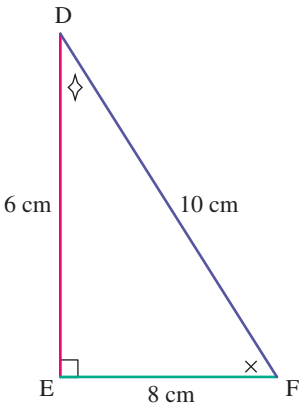
5.2.1 Similar right-angled triangles

eles-4799

- Two similar right-angled triangles have the same angles when the corresponding sides are in the same ratio.
- The **hypotenuse** is the longest side of a right-angled triangle and is always the side that is opposite the right angle.
- The corresponding sides are in the same ratio.

$$\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$



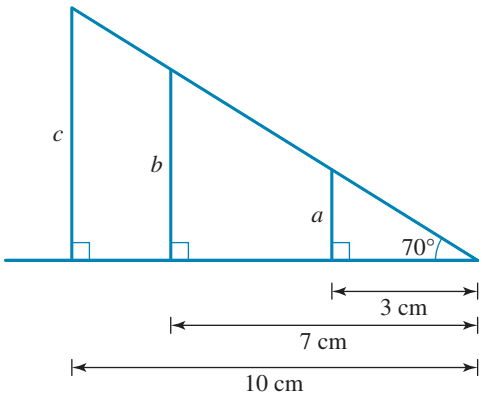


- To write this using the side lengths of the triangles gives:

$$\frac{AB}{DE} = \frac{3}{6} = \frac{1}{2}$$
$$\frac{AC}{DF} = \frac{5}{10} = \frac{1}{2}$$
$$\frac{BC}{EF} = \frac{4}{8} = \frac{1}{2}$$

This means that for right-angled triangles, when the angles are fixed, the ratios of the sides in the triangle are constant.

- We can examine this idea further by completing the following activity.
  Using a protractor and ruler, draw an angle of 70° measuring horizontal distances of 3 cm, 7 cm and 10 cm as demonstrated in the diagram below.



Note: Diagram not drawn to scale.

Measure the perpendicular heights a, b and c.

$$a \approx 8.24 \text{ cm}, \ b \approx 19.23 \text{ cm}, \ c \approx 27.47 \text{ cm}$$

- To test if the theory for right-angled triangles, that when the angles are fixed the ratios of the sides in the triangle are constant, is correct, calculate the ratios of the side lengths.

$$\frac{a}{3} \approx \frac{8.24}{3} \approx 2.75, \quad \frac{b}{7} \approx \frac{19.23}{7} \approx 2.75, \quad \frac{c}{10} \approx \frac{27.47}{10} \approx 2.75$$

The ratios are the same because the triangles are similar. This important concept forms the basis of trigonometry.

5.2.2 Review of Pythagoras' theorem

eles-4800

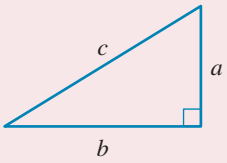
- Pythagoras' theorem gives us a way of calculating the length of the third side in a right angle triangle, if we know the lengths of the two other sides.

Pythagoras' theorem

Pythagoras' theorem: $a^2 + b^2 = c^2$

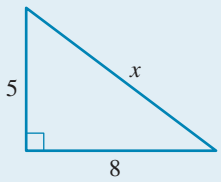
Determining the hypotenuse: $c = \sqrt{a^2 + b^2}$

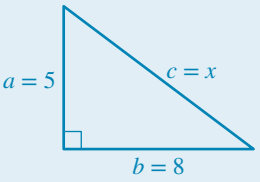
Determining one of the two shorter sides: $a = \sqrt{c^2 - b^2}$ **or** $b = \sqrt{c^2 - a^2}$



WORKED EXAMPLE 1 Calculating the hypotenuse

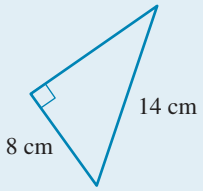
For the following triangle, calculate the length of the hypotenuse x, correct to 1 decimal place.



THINK	WRITE/DRAW
1. Copy the diagram and label the sides a, b and c. Remember to label the hypotenuse as c.	
2. Write Pythagoras' theorem.	$c^2 = a^2 + b^2$
3. Substitute the values of a, b and c into this rule and simplify.	$x^2 = 5^2 + 8^2$ $= 25 + 64$ $= 89$
4. Take the square root of both sides. Round the positive answer correct to 1 decimal place, since $x > 0$.	$x = \pm\sqrt{89}$ $x \approx 9.4$

WORKED EXAMPLE 2 Calculating the length of the shorter side

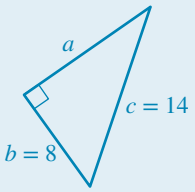
Calculate the length, correct to 1 decimal place, of the unmarked side of the following triangle.



THINK

1. Copy the diagram and label the sides a, b and c. Remember to label the hypotenuse as c; it does not matter which side is a and which side is b.

WRITE/DRAW



2. Write Pythagoras' theorem.

$c^2 = a^2 + b^2$

3. Substitute the values of a, b and c into this rule and solve for a.

$14^2 = a^2 + 8^2$

$196 = a^2 + 64$

$a^2 = 196 - 64$

$= 132$

4. Evaluate a by taking the square root of both sides and round to 1 decimal place ($a > 0$).

$a = \pm\sqrt{132}$

≈ 11.5 cm

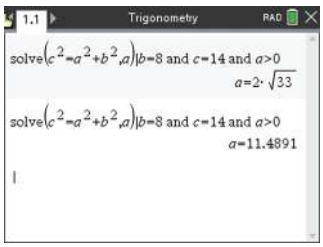
TI | THINK

In a new document, on a Calculator page, to solve equations press:

- MENU
- 3: Algebra
- 1: Solve

Complete the entry line as:
solve $\left(c^2 = a^2 + b^2, a\right)$
$|b = 8$ and $c = 14$ and $a > 0$
Then press ENTER.
Press CTRL ENTER to get a decimal approximation.

DISPLAY/WRITE

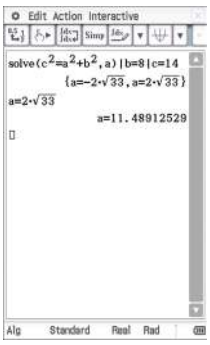


The length of the unmarked side is $a = 2\sqrt{33} = 11.5$ correct to 1 decimal place.

CASIO | THINK

On the Main screen, complete the entry line as:
solve $\left(c^2 = a^2 + b^2, a\right)$
$|b = 8|\, c = 14$
Then press EXE.
To convert to decimals, highlight the answer that is greater than 0 and drag it to a new line. Change the mode to decimal then press EXE.

DISPLAY/WRITE



The length of the unmarked side is $a = 2\sqrt{33} = 11.5$ correct to 1 decimal place.

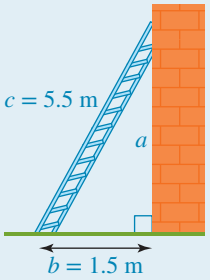
WORKED EXAMPLE 3 Solving a practical problem using Pythagoras' theorem

A ladder that is 5.5 m long leans up against a vertical wall. The foot of the ladder is 1.5 m from the wall. Determine how far up the wall the ladder reaches. Give your answer in metres correct to 1 decimal place.

THINK

1. Draw a diagram and label the sides a, b and c. Remember to label the hypotenuse as c.

WRITE/DRAW



2. Write Pythagoras' theorem.

$$c^2 = a^2 + b^2$$

3. Substitute the values of a, b and c into this rule and simplify.

$$5.5^2 = a^2 + 1.5^2$$
$$30.25 = a^2 + 2.25$$
$$a^2 = 30.25 - 2.25$$
$$= 28$$

4. Evaluate a by taking the square root of 28. Round to 1 decimal place, $a > 0$.

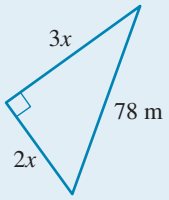
$$a = \pm\sqrt{28}$$
$$\approx 5.3$$

5. Write the answer in a sentence.

The ladder reaches 5.3 m up the wall.

WORKED EXAMPLE 4 Determining the unknown sides

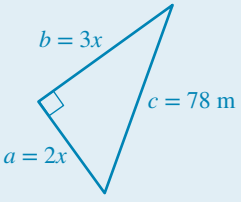
Determine the unknown side lengths of the triangle, correct to 2 decimal places.



THINK

1. Copy the diagram and label the sides a, b and c.

WRITE/DRAW



2. Write Pythagoras' theorem.

$$c^2 = a^2 + b^2$$

3. Substitute the values of a, b and c into this rule and simplify.

$$78^2 = (3x)^2 + (2x)^2$$
$$6084 = 9x^2 + 4x^2$$
$$6084 = 13x^2$$

4. Rearrange the equation so that the pronumeral is on the left-hand side of the equation.
$13x^2 = 6084$

5. Divide both sides of the equation by 13.
$$\frac{13x^2}{13} = \frac{6084}{13}$$
$$x^2 = 468$$

6. Evaluate x by taking the square root of both sides. Round the answer correct to 2 decimal places.
$$x = \pm\sqrt{468}$$
$$\approx 21.6333$$

7. Substitute the value of x into $2x$ and $3x$ to determine the lengths of the unknown sides.
$2x \approx 43.27\text{m}$
$3x \approx 64.90\text{ m}$

DISCUSSION

Pythagoras' theorem was known about before the age of Pythagoras. Research which other civilisations knew about the theory and construct a timeline for its history.

on Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Interactivities Individual pathway interactivity: Pythagoras' theorem (int-4585)
Finding the hypotenuse (int-3844)
Finding a shorter side (int-3845)

Exercise 5.2 Pythagoras' theorem

learnon

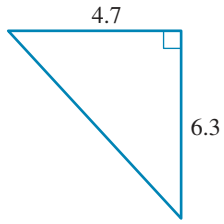
5.2 Quick quiz on | **5.2 Exercise**

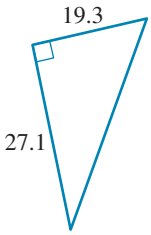
Individual pathways

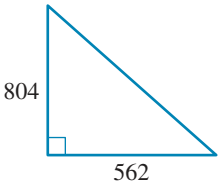
PRACTISE	CONSOLIDATE	MASTER
1, 5, 7, 10, 11, 14, 17, 21, 22, 25	2, 3, 8, 12, 15, 19, 20, 23, 26	4, 6, 9, 13, 16, 18, 24, 27

Fluency

1. WE1 For each of the following triangles, calculate the length of the hypotenuse, giving answers correct to 2 decimal places.

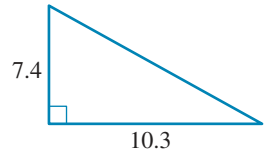
a.


b.


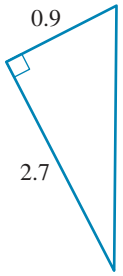
c.


2. For each of the following triangles, calculate the length of the hypotenuse, giving answers correct to 2 decimal places.

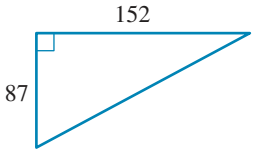
a.



b.

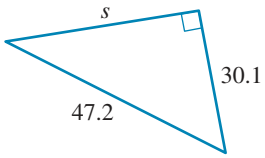


c.

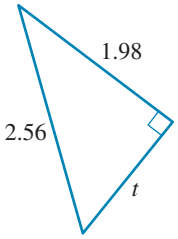


3. **WE2** Determine the value of the pronumeral, correct to 2 decimal places.

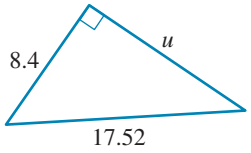
a.



b.

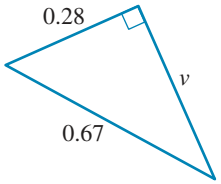


c.

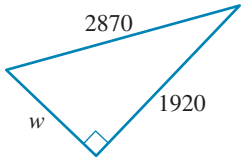


4. Determine the value of the pronumeral, correct to 2 decimal places.

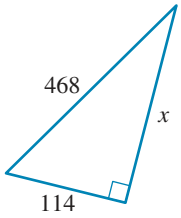
a.



b.



c.



5. **WE3** The diagonal of the rectangular NO SMOKING sign is 34 cm. If the height of this sign is 25 cm, calculate the width of the sign, in cm correct to 2 decimal places.



6. A right-angled triangle has a base of 4 cm and a height of 12 cm. Calculate the length of the hypotenuse in cm correct to 2 decimal places.

7. Calculate the lengths of the diagonals (in cm to 2 decimal places) of squares that have side lengths of:

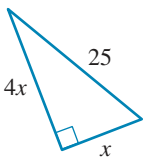
a. 10 cm **b.** 17 cm **c.** 3.2 cm.

8. The diagonal of a rectangle is 90 cm. One side has a length of 50 cm. Determine, correct to 2 decimal places:

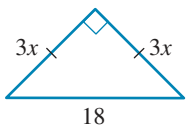
a. the length of the other side
b. the perimeter of the rectangle
c. the area of the rectangle.

9. **WE4** Determine the value of the pronumeral, correct to 2 decimal places for each of the following.

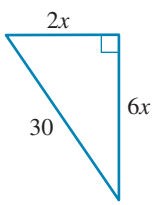
a.



b.



c.



Understanding

10. An isosceles triangle has a base of 25 cm and a height of 8 cm. Calculate the length of the two equal sides, in cm correct to 2 decimal places.

11. An equilateral triangle has sides of length 18 cm. Determine the height of the triangle, in cm correct to 2 decimal places.

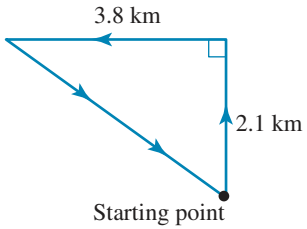
12. A right-angled triangle has a height of 17.2 cm, and a base that is half the height. Calculate the length of the hypotenuse, in cm correct to 2 decimal places.

13. The road sign shown is based on an equilateral triangle. Determine the height of the sign and, hence, calculate its area. Round your answers to 2 decimal places.

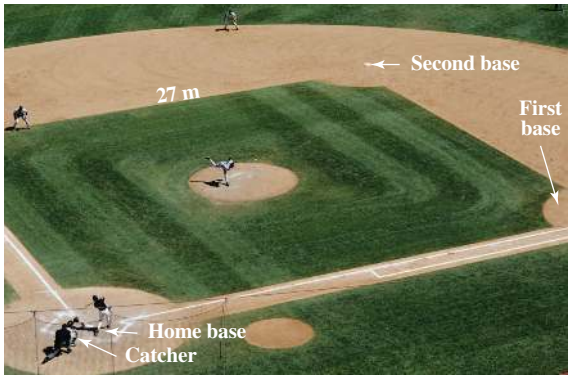


14. A flagpole, 12 m high, is supported by three wires, attached from the top of the pole to the ground. Each wire is pegged into the ground 5 m from the pole. Determine how much wire is needed to support the pole, correct to the nearest metre.

15. Sarah goes canoeing in a large lake. She paddles 2.1 km to the north, then 3.8 km to the west. Use the triangle shown to determine how far she must then paddle to get back to her starting point in the shortest possible way, in km correct to 2 decimal places.



16. A baseball diamond is a square of side length 27 m. When a runner on first base tries to steal second base, the catcher has to throw the ball from home base to second base. Calculate the distance of the throw, in metres correct to 1 decimal place.



17. A rectangle measures 56 mm by 2.9 cm. Calculate the length of its diagonal in millimetres correct to 2 decimal places.

18. A rectangular envelope has a length of 24 cm and a diagonal measuring 40 cm. Calculate:

a. the width of the envelope, correct to the nearest cm

b. the area of the envelope, correct to the nearest cm^2.

19. A swimming pool is 50 m by 25 m. Peter is bored by his usual training routine, and decides to swim the diagonal of the pool. Determine how many diagonals he must swim to complete his normal distance of 1500 m.

20. A hiker walks 2.9 km north, then 3.7 km east. Determine how far in metres she is from her starting point. Give your answer in metres to 2 decimal places.

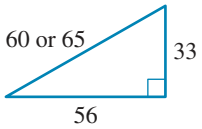


21. A square has a diagonal of 14 cm. Calculate the length of each side, in cm correct to 2 decimal places.

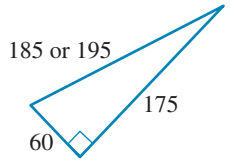
Reasoning

22. The triangles below are right-angled triangles. Two possible measurements have been suggested for the hypotenuse in each case. For each triangle, complete calculations to determine which of the lengths is correct for the hypotenuse in each case.
Show your working.

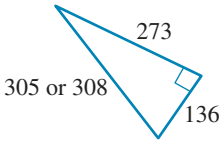
a.



b.



c.



23. The square root of a number usually gives us both a positive and negative answer. Explain why we take only the positive answer when using Pythagoras' theorem.

24. Four possible side length measurements are 105, 208, 230 and 233. Three of them together produce a right-angled triangle.

a. Explain which of the measurements could not be the hypotenuse of the triangle.

b. Complete as few calculations as possible to calculate which combination of side lengths will produce a right-angled triangle.

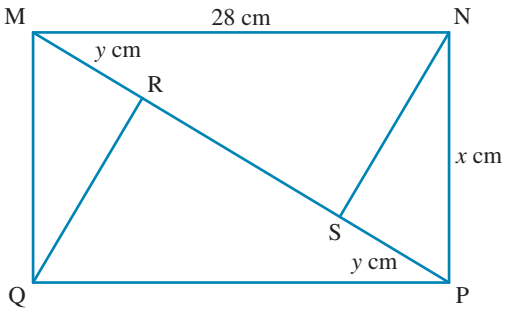
Problem solving

25. The area of the rectangle MNPQ is 588 cm^2. Angles MRQ and NSP are right angles.

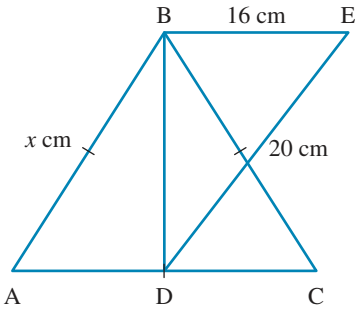
a. Determine the integer value of x.

b. Determine the length of MP.

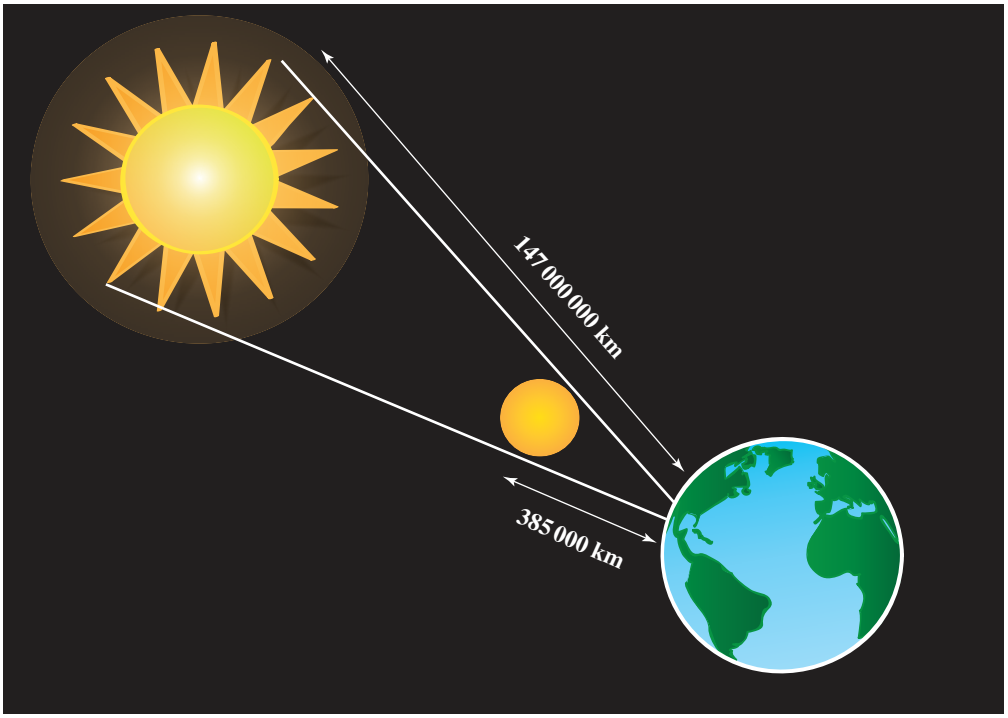
c. Calculate the value of y and hence determine the length of RS, in cm correct to 1 decimal place.



26. Triangle ABC is an equilateral triangle of side length x cm. Angles ADB and DBE are right angles. Determine the value of x in cm, correct to 2 decimal places.



27. The distance from Earth to the Moon is approximately 385 000 km and the distance from Earth to the Sun is approximately 147 million kilometres.
In a total eclipse of the Sun, the moon moves between the Sun and Earth, thus blocking the light of the Sun from reaching Earth and causing a total eclipse of the Sun.
If the diameter of the Moon is approximately 3474 km, evaluate the diameter of the Sun. Express your answer to the nearest 10 000 km.



LESSON
5.3 Pythagoras' theorem in three dimensions (10A)

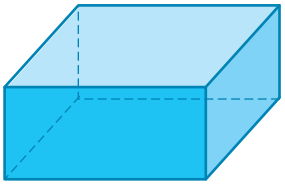
LEARNING INTENTIONS

At the end of this lesson you should be able to apply Pythagoras' theorem to:
- determine unknown lengths when a 3D diagram is given
- determine unknown lengths in situations by first drawing a diagram.

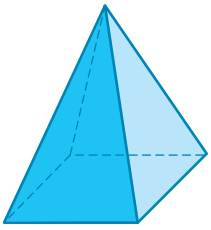
eles-4801

5.3.1 Applying Pythagoras' theorem in three dimensions

- Many real-life situations involve **3-dimensional** (3D) objects: objects with length, width and height. Some common 3D objects used in this section include cuboids, pyramids and right-angled wedges.



Cuboid



Pyramid

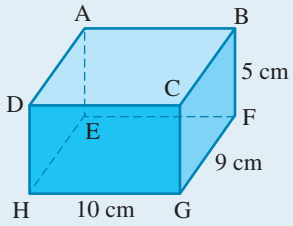


Right-angled wedge

- In diagrams of 3D objects, right angles may not look like right angles, so it is important to redraw sections of the diagram in two dimensions, where the right angles can be seen accurately.

WORKED EXAMPLE 5 Applying Pythagoras' theorem to 3D objects

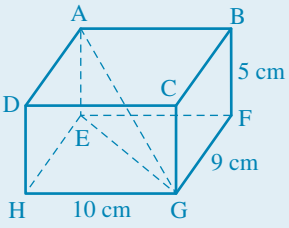
Determine the length AG in this rectangular prism (cuboid), in cm correct to 2 decimal places.



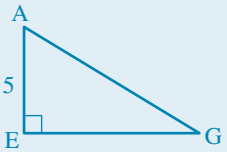
THINK

1. Draw the diagram in three dimensions.
   Draw the lines AG and EG.
   ∠AEG is a right angle.

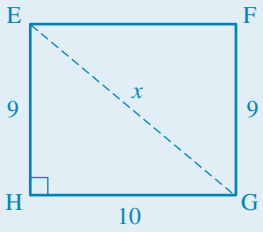
WRITE/DRAW



2. Draw ΔAEG, showing the right angle. Only 1 side is known, so EG must be found.



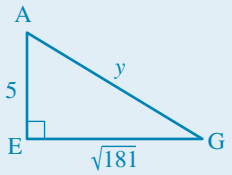
3. Draw EFGH in two dimensions and label the diagonal EG as x.



4. Use Pythagoras' theorem to calculate x. $\left(c^2 = a^2 + b^2\right)$

$$\begin{aligned} x^2 &= 9^2 + 10^2 \\ &= 81 + 100 \\ &= 181 \\ x &= \sqrt{181} \end{aligned}$$

5. Place this information on triangle AEG. Label the side AG as y.



6. Use Pythagoras' theorem to calculate y. $\left(c^2 = a^2 + b^2\right)$

$$\begin{aligned} y^2 &= 5^2 + \left(\sqrt{181}\right)^2 \\ &= 25 + 181 \\ &= 206 \\ y &= \sqrt{206} \\ &\approx 14.35 \end{aligned}$$

7. Write the answer in a sentence.

The length of AG is 14.35 cm.

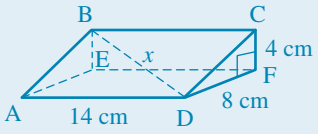
WORKED EXAMPLE 6 Drawing a diagram to solve problems

A piece of cheese in the shape of a right-angled wedge sits on a table. It has a rectangular base measuring 14 cm by 8 cm, and is 4 cm high at the thickest point. An ant crawls diagonally across the sloping face.
Determine how far, to the nearest millimetre, the ant walks.

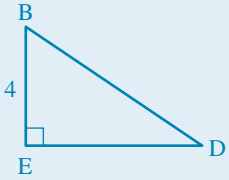
THINK

1. Draw a diagram in three dimensions and label the vertices. Mark BD, the path taken by the ant, with a dotted line. ∠BED is a right angle.

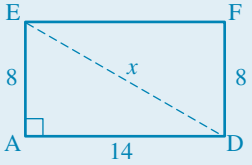
WRITE/DRAW



2. Draw ΔBED, showing the right angle. Only one side is known, so ED must be found.



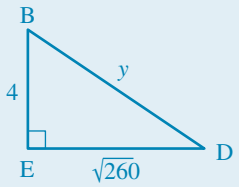
3. Draw EFDA in two dimensions and label the diagonal ED. Label the side ED as x.



4. Use Pythagoras' theorem to calculate x.

$$c^2 = a^2 + b^2$$
$$x^2 = 8^2 + 14^2$$
$$= 64 + 196$$
$$= 260$$
$$x = \sqrt{260}$$

5. Place this information on triangle BED. Label the side BD as y.



6. Use Pythagoras' theorem to calculate y.

$$y^2 = 4^2 + \left(\sqrt{260}\right)^2$$
$$= 16 + 260$$
$$= 276$$
$$y = \sqrt{276}$$
$$\approx 16.61 \text{ cm}$$
$$\approx 166.1 \text{ mm}$$

7. Write the answer in a sentence.

The ant walks 166 mm, correct to the nearest millimetre.

DISCUSSION

Look around the room you are in. How many right angles you can spot in three-dimensional objects? Make a list of them and compare your list to that of another student.

on Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Video eLesson Pythagoras' theorem in three dimensions (eles-1913)

Interactivity Right angles in 3-dimensional objects (int-6132)

Exercise 5.3 Pythagoras' theorem in three dimensions (10A)

learnon

5.3 Quick quiz on | 5.3 Exercise

Individual pathways

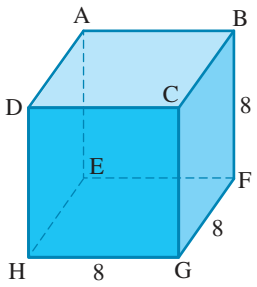
PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 11, 12, 15	2, 5, 8, 13, 16	3, 6, 9, 10, 14, 17

Where appropriate in this exercise, give answers correct to 2 decimal places.

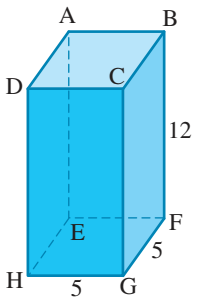
Fluency

1. WE5 Calculate the length of AG in each of the following figures.

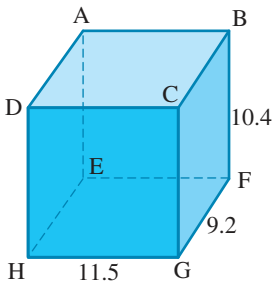
a.



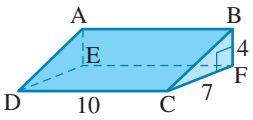
b.



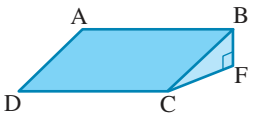
c.



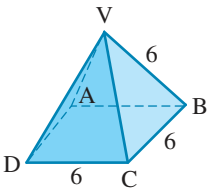
2. Consider the wedge shown. Calculate the length of CE in the wedge and, hence, obtain the length of AC .



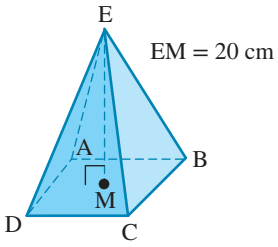
3. If DC = 3.2 m, AC = 5.8 m, and CF = 4.5 m in the figure, calculate the lengths of AD and BF.



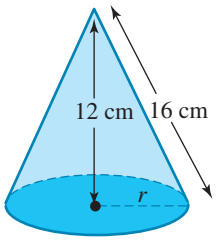
4. Consider the pyramid shown. Calculate the length of BD and, hence, the height of the pyramid.



5. The pyramid ABCDE has a square base. The pyramid is 20 cm high. Each sloping edge measures 30 cm. Calculate the length of the sides of the base.

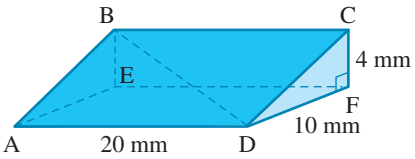


6. The sloping side of a cone is 16 cm and the height is 12 cm. Determine the length of the radius of the base.

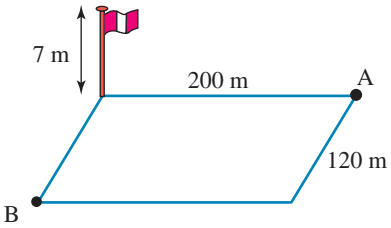


Understanding

7. **WE6** A piece of cheese in the shape of a right-angled wedge sits on a table. It has a base measuring 20 mm by 10 mm, and is 4 mm high at the thickest point, as shown in the figure. A fly crawls diagonally across the sloping face. Determine how far, to the nearest millimetre, the fly walks.



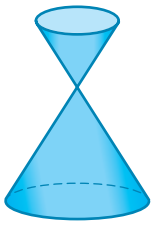
8. A 7 m high flagpole is in the corner of a rectangular park that measures 200 m by 120 m. Give your answers to the following questions correct to 2 decimal places.



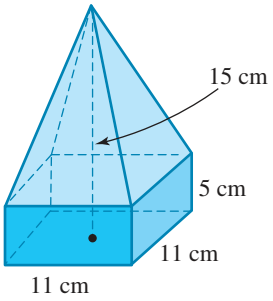
   a. Calculate:
      i. the length of the diagonal of the park
      ii. the distance from A to the top of the pole
      iii. the distance from B to the top of the pole.

   b. A bird flies from the top of the pole to the centre of the park. Calculate how far it flies.

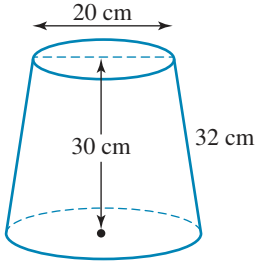
9. A candlestick is in the shape of two cones, joined at the vertices as shown. The smaller cone has a diameter and sloping side of 7 cm, and the larger one has a diameter and sloping side of 10 cm. Calculate the total height of the candlestick.



10. The total height of the shape below is 15 cm. Calculate the length of the sloping side of the pyramid.



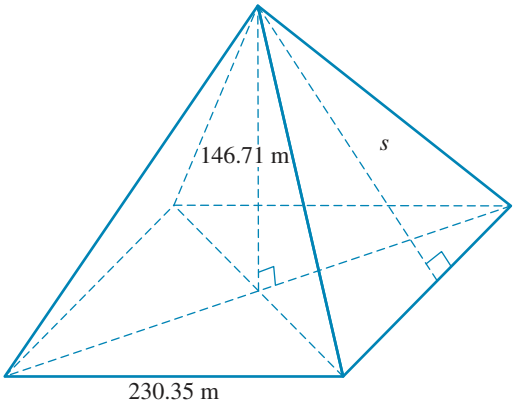
11. A sandcastle is in the shape of a truncated cone as shown. Calculate the length of the diameter of the base.



Reasoning

12. Stephano is renovating his apartment, which he accesses through two corridors. The corridors of the apartment building are 2 m wide with 2 m high ceilings, and the first corridor is at right angles to the second. Show that he can carry lengths of timber up to 6 m long to his apartment.

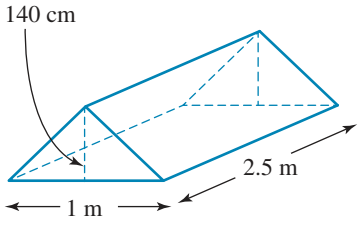
13. The Great Pyramid in Egypt is a square-based pyramid. The square base has a side length of 230.35 metres and the perpendicular height is 146.71 metres.
Determine the slant height, s, of the great pyramid. Give your answer correct to 1 decimal place.



14. A tent is in the shape of a triangular prism, with a height of 140 cm as shown in the diagram. The width across the base of the door is 1 m and the tent is 2.5 m long.

a. Calculate the length of each sloping side, in metres.

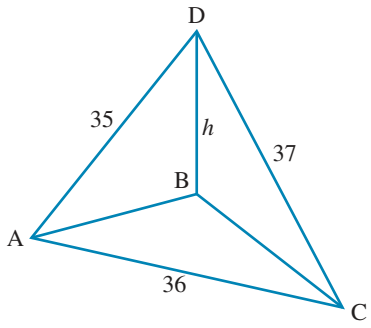
b. Using your answer from part **a** calculate the area of fabric used in the construction of the sloping rectangles which form the sides. Show full working.



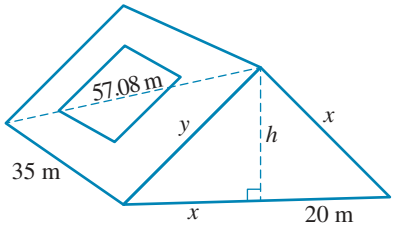
Problem solving

15. Determine the exact length of the longest steel rod that can sit inside a cuboid with dimensions 32 cm × 15 cm × 4 cm. Ignore the thickness of the steel rod.

16. Angles ABD, CBD and ABC are right angles. Determine the value of h, correct to 3 decimal places.



17. The roof of a squash centre is constructed to allow for maximum use of sunlight. Determine the value of h, giving your answer correct to 1 decimal place.



LESSON
5.4 Trigonometric ratios

LEARNING INTENTIONS

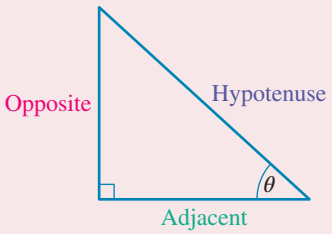
At the end of this lesson you should be able to:
- define trigonometric ratios according to the lengths of the relevant sides
- write equations for trigonometric ratios.

5.4.1 Trigonometric ratios

eles-4802

- In a right-angled triangle, the longest side is called the hypotenuse.
- If one of the two acute angles is named (for example, θ), then the other two sides can also be given names, as shown in the following diagram.

Three basic definitions



- **Using the diagram, the following three trigonometric ratios can be defined.**
  - **The sine ratio:**

$$\text{sine}(\theta) = \frac{\text{length of Opposite side}}{\text{length of Hypotenuse}}$$

  - **The cosine ratio:**

$$\text{cosine}(\theta) = \frac{\text{length of Adjacent side}}{\text{length of Hypotenuse}}$$

  - **The tangent ratio:**

$$\text{tangent}(\theta) = \frac{\text{length of Opposite side}}{\text{length of Adjacent side}}$$

- **The names of the three ratios are usually shortened to sin(θ), cos(θ) and tan(θ).**
- **The three ratios are often remembered using the mnemonic SOHCAHTOA, where SOH means Sin(θ) = Opposite over Hypotenuse and so on.**

Calculating trigonometric values using a calculator

- The sine, cosine and tangent of an angle have numerical values that can be found using a calculator.
- Traditionally angles were measured in **degrees**, minutes and seconds, where 60 seconds = 1 minute and 60 minutes = 1 degree. This is known as a sexagesimal system as the division are based on 60.
  For example, 50°33′48″ means 50 degrees, 33 minutes and 48 seconds.

WORKED EXAMPLE 7 Calculating values (ratios) from angles

Calculate the value of each of the following, correct to 4 decimal places, using a calculator. (Remember to first work to 5 decimal places before rounding.)

a. $\cos(65°57')$ **b.** $\tan(56°45'30'')$

THINK	WRITE
a. Write your answer to the required number of decimal places.	**a.** $\cos(65°57') \approx 0.40753$ ≈ 0.4075
b. Write your answer to the correct number of decimal places.	**b.** $\tan(56°45'30'') \approx 1.52573$ ≈ 1.5257

TI | THINK

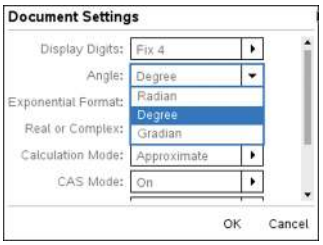
a, b.

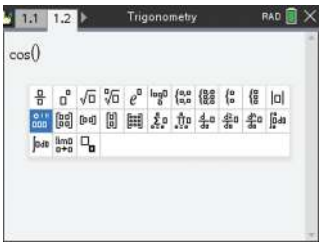
1. To ensure your calculator is set to degree and approximate mode, press:
   - HOME
   - 5: Settings
   - 2: Document Settings

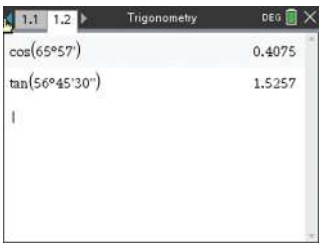
   In the Display Digits, select Fix 4. Tab to Angle and select Degree; tab to Calculation Mode and select Approximate.
   Tab to OK and press ENTER
2. On a Calculator page, press TRIG to access and select the appropriate trigonometric ratio. Then press ⇨ and choose the template for degrees, minutes and seconds as shown.
3. Complete the entry lines as:
   $\cos(65°57')$
   $\tan(56°45'30'')$
   Press ENTER after each entry.
   Since the Calculation Mode is set to Approximate and Fix 4, the answer are shown correct to 4 decimal places.

DISPLAY/WRITE

a, b.







$\cos(65°57') = 0.4075$
$\tan(56°45'30'') = 1.5257$

CASIO | THINK

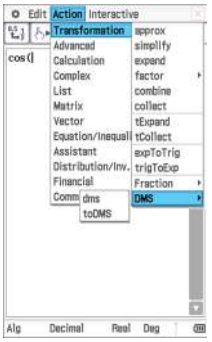
a, b.

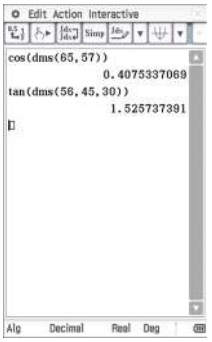
1. Ensure your calculator is set to degrees at the bottom right. If not, tap 'Rad' or 'Gra'.
   Tap unit Deg appears.
   Set to Decimal.
   Set the keyboard to Trig.
2. Complete the entry line as:
   cos(dms(65, 57))
   tan(dms(56, 45, 30))
   To get dms, tap
   - Action
   - Transformation
   - DMS
   - dms

   Then press EXE after each entry.

DISPLAY/WRITE

a, b.





$\cos(65°57') = 0.4075$
$\tan(56°45'30'') = 1.5257$

WORKED EXAMPLE 8 Calculating angles from ratios

Calculate the size of angle θ, correct to the nearest degree, given $\sin(\theta) = 0.7854$.

THINK	WRITE
1. Write the given equation.	$\sin(\theta) = 0.7854$
2. To calculate the size of the angle, we need to undo sine with its inverse, $\sin^{-1}$. (Ensure your calculator is in degrees mode.)	$\theta = \sin^{-1}(0.7854)$ $\approx 51.8°$
3. Write your answer to the nearest degree.	$\theta \approx 52°$

WORKED EXAMPLE 9 Expressing angles in degrees, minutes and seconds

Calculate the value of θ:
a. correct to the nearest minute, given that $\cos(\theta) = 0.2547$
b. correct to the nearest second, given that $\tan(\theta) = 2.364$.

THINK	WRITE
a. 1. Write the equation.	a. $\cos(\theta) = 0.2547$
2. Write your answer, including seconds. There are 60 seconds in 1 minute. Round to the nearest minute. (Remember $60'' = 1'$, so $39''$ is rounded up.)	$\cos^{-1}(0.2547) \approx 75°14'39''$ $\approx 75°15'$
b. 1. Write the equation.	b. $\tan(\theta) = 2.364$
2. Write the answer, rounding to the nearest second.	$\tan^{-1}(2.364) \approx 67°4'15.8''$ $\approx 67°4'16''$

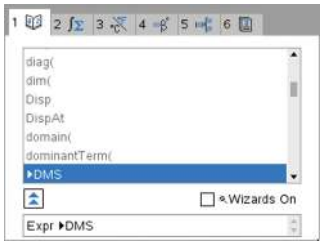
TI | THINK

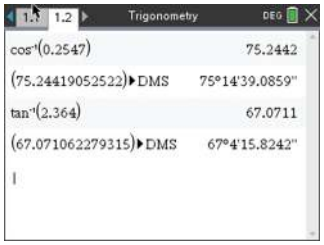
a, b.
On a Calculator page, press TRIG to access and select the appropriate trigonometric ratio, in the case $\cos^{-1}$.
Complete the entry lines as:
$\cos^{-1}(0.2547)$
To convert the decimal degree into degrees, minutes and seconds, press:
- CATALOG
- 1
- D

Scroll and select ▶ DMS
Then press ENTER.
Repeat this process for $\tan^{-1}(2.364)$.

DISPLAY/WRITE

a, b.





$\cos^{-1}(0.2547) = 75°15'$ rounding to the nearest minute.
$\tan^{-1}(2.364) = 64°4'16''$ rounding to the nearest second.

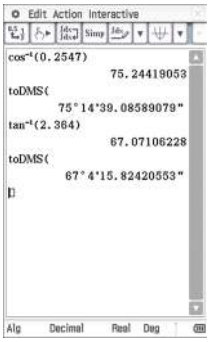
CASIO | THINK

a, b.
Ensure your calculator is set to degrees at the bottom right.
On the Main screen in decimal and degree mode, complete the entry line as:
$\cos^{-1}(0.2547)$
Then press EXE.
To convert the decimal answer into degrees, minutes and seconds, tap:
- Action
- Transformation
- DMS
- to DMS

Then press EXE.
Repeat this process for $\tan^{-1}(2.364)$.

DISPLAY/WRITE

a, b.

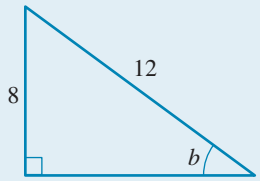


$\cos^{-1}(0.2547) = 75°15'$ rounding to the nearest minute.
$\tan^{-1}(2.364) = 64°4'16''$ rounding to the nearest second.

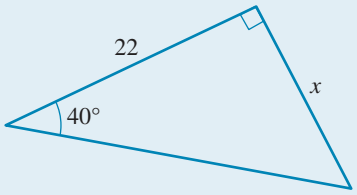
WORKED EXAMPLE 10 Expressing trigonometric ratios as equations

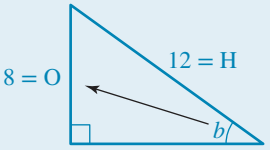
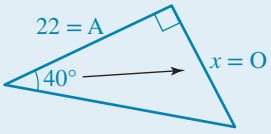
Write the equation that relates the two marked sides and the marked angle.

a.



b.



THINK	WRITE/DRAW
a. 1. Label the given sides of the triangle.	**a.** 
2. Write the ratio that contains O and H.	$\sin(\theta) = \frac{\text{O}}{\text{H}}$
3. Identify the values of the pronumerals.	$\text{O} = 8, \text{H} = 12$
4. Substitute the values of the pronumerals into the ratio and simplify the fraction. (Since the given angle is denoted with the letter b, replace θ with b.)	$\sin(b) = \frac{8}{12} = \frac{2}{3}$
b. 1. Label the given sides of the triangle.	**b.** 
2. Write the ratio that contains O and A.	$\tan(\theta) = \frac{\text{O}}{\text{A}}$
3. Identify the values of the pronumerals.	$\text{O} = x, \text{A} = 22, \theta = 40°$
4. Substitute the values of the pronumerals into the ratio.	$\tan(40°) = \frac{x}{22}$

DISCUSSION

Do you know of any other mnemonics that you can use to help you remember important information?

on Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Interactivities Individual pathway interactivity: Trigonometric ratios (int-4587)
Trigonometric ratios (int-2577)

Exercise 5.4 Trigonometric ratios

learnon

5.4 Quick quiz on | 5.4 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 5, 7, 9, 11, 14, 15, 22, 25	2, 6, 12, 16, 17, 20, 23, 26, 27	3, 4, 8, 10, 13, 18, 19, 21, 24, 28

Fluency

1. Calculate each of the following, correct to 4 decimal places.

a. $\sin(30°)$ **b.** $\cos(45°)$ **c.** $\tan(25°)$
d. $\sin(57°)$ **e.** $\tan(83°)$ **f.** $\cos(44°)$

WE7 For questions **2** to **4**, calculate each of the following, correct to 4 decimal places.

2. **a.** $\sin(40°30')$ **b.** $\cos(53°57')$ **c.** $\tan(27°34')$
d. $\tan(123°40')$ **e.** $\sin(92°32')$ **f.** $\sin(42°8')$

3. **a.** $\cos(35°42'35'')$ **b.** $\tan(27°42'50'')$ **c.** $\cos(143°25'23'')$
d. $\sin(23°58'21'')$ **e.** $\cos(8°54'2'')$

4. **a.** $\sin(286)°$ **b.** $\tan(420°)$ **c.** $\cos(845°)$ **d.** $\sin(367°35')$

5. **WE8** Calculate the size of angle θ, correct to the nearest degree, for each of the following.

a. $\sin(\theta) = 0.763$ **b.** $\cos(\theta) = 0.912$ **c.** $\tan(\theta) = 1.351$

6. Calculate the size of angle θ, correct to the nearest degree, for each of the following.

a. $\cos(\theta) = 0.321$ **b.** $\tan(\theta) = 12.86$ **c.** $\cos(\theta) = 0.756$

7. **WE9a** Calculate the size of the angle θ, correct to the nearest minute.

a. $\sin(\theta) = 0.814$ **b.** $\sin(\theta) = 0.110$ **c.** $\tan(\theta) = 0.015$

8. Calculate the size of the angle θ, correct to the nearest minute.

a. $\cos(\theta) = 0.296$ **b.** $\tan(\theta) = 0.993$ **c.** $\sin(\theta) = 0.450$

9. **WE9b** Calculate the size of the angle θ, correct to the nearest second.

a. $\tan(\theta) = 0.5$ **b.** $\cos(\theta) = 0.438$ **c.** $\sin(\theta) = 0.9047$

10. Calculate the size of the angle θ, correct to the nearest second.

a. $\tan(\theta) = 1.1141$ **b.** $\cos(\theta) = 0.8$ **c.** $\tan(\theta) = 43.76.$

For questions **11** to **13**, calculate the value of each expression, correct to 3 decimal places.

11. **a.** $3.8\ \cos(42°)$ **b.** $118\ \sin(37°)$ **c.** $2.5\ \tan(83°)$ **d.** $\dfrac{2}{\sin(45°)}$

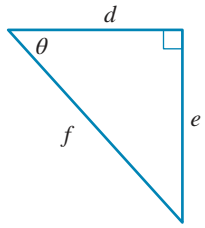
12. **a.** $\dfrac{220}{\cos(14°)}$ **b.** $\dfrac{2\ \cos(23°)}{5\ \sin(18°)}$ **c.** $\dfrac{12.8}{\tan(60°32')}$ **d.** $\dfrac{18.7}{\sin(35°25'42'')}$

13. **a.** $\dfrac{55.7}{\cos(89°21')}$ **b.** $\dfrac{3.8\tan(1°51'44'')}{4.5\sin(25°45')}$ **c.** $\dfrac{2.5\sin(27°8')}{10.4\cos(83°2')}$ **d.** $\dfrac{3.2\cos(34°52')}{0.8\sin(12°48')}$

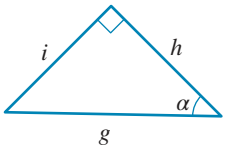
For questions 14 to 19, write an expression for:

a. sine **b.** cosine **c.** tangent.

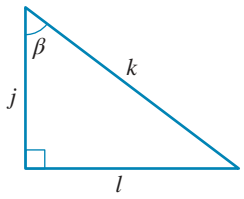
14.



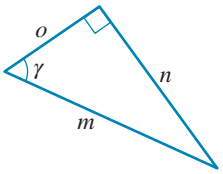
15.



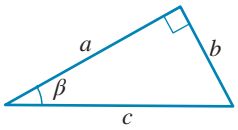
16.



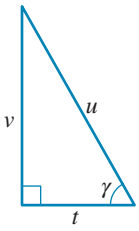
17.



18.



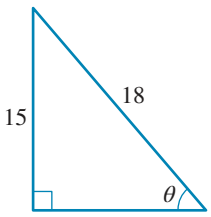
19.



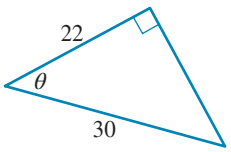
Understanding

20. WE10 Write the equation that relates the two marked sides and the marked angle in each of the following triangles.

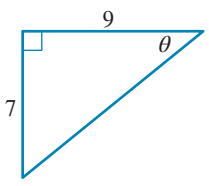
a.



b.

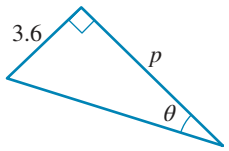


c.

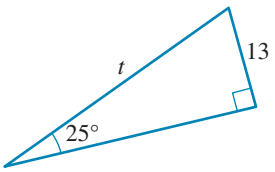


21. Write the equation that relates the two marked sides and the marked angle in each of the following triangles.

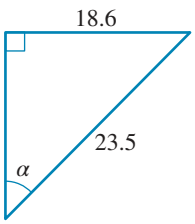
a.



b.

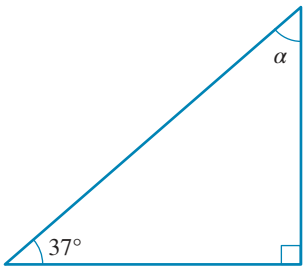


c.



Reasoning

22. Consider the right-angled triangle shown.



a. Label each of the sides using the letters O, A and H with respect to the 37° angle.

b. Determine the value of each trigonometric ratio. (Where applicable, answers should be given correct to 2 decimal places.)

i. sin(37°) **ii.** cos(37°) **iii.** tan(37°)

c. Determine the value of the unknown angle, α.

23. Consider the right-angled triangle shown in question **22**.

a. Determine the value of each of these trigonometric ratios, correct to 2 decimal places.

i. sin(α) **ii.** cos(α) **iii.** tan(α)

(*Hint:* First relabel the sides of the triangle with respect to angle α.)

b. What do you notice about the relationship between sin(37°) and cos(α)? Explain your answer.

c. What do you notice about the relationship between sin(α) and cos(37°)? Explain your answer.

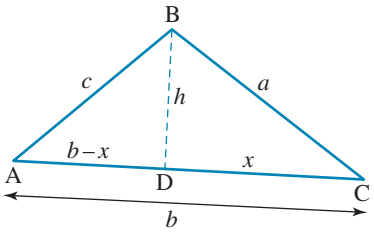
d. Make a general statement about the two angles.

24. Using a triangle labelled with a, h and o, algebraically show that $\tan(\theta) = \dfrac{\sin(\theta)}{\cos(\theta)}$.

(*Hint:* Write all the sides in terms of the hypotenuse.)

Problem solving

25. ABC is a scalene triangle with side lengths a, b and c as shown. Angles BDA and BDC are right angles.



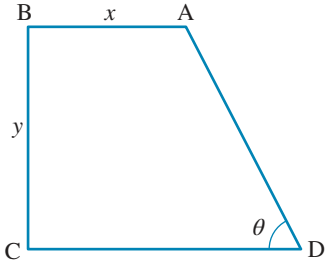
a. Express h^2 in terms of a and x.

b. Express h^2 in terms of b, c and x.

c. Equate the two equations for h^2 to show that $c^2 = a^2 + b^2 - 2bx$.

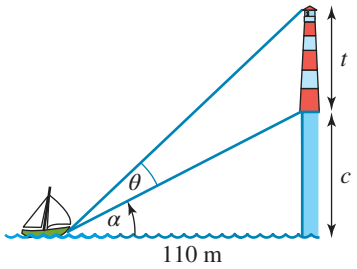
d. Use your knowledge of trigonometry to produce the equation $c^2 = a^2 + b^2 - 2ab\cos(C)$, which is known as the cosine rule for non-right-angled triangles.

26. Determine the length of the side DC in terms of x, y and θ.



27. Explain how we determine whether to use sin, cos or tan in trigonometry questions.

28. From an observer on a boat 110 m away from a vertical cliff with height c, the angle from the base of the cliff to the top of the cliff is $\alpha°$. There is a lighthouse with height t on the cliff. From the observer, the angle from the base of the lighthouse to the top of the lighthouse is another $\theta°$ more than $\alpha°$.
Express the height of the lighthouse, t, in terms of $\theta°$ and $\alpha°$.



LESSON
5.5 Using trigonometry to calculate side lengths

LEARNING INTENTION

At the end of this lesson you should be able to:

- apply trigonometric ratios to calculate the length of an unknown side when the length of one other side and an acute angle is known.

5.5.1 Using trigonometry to calculate side lengths

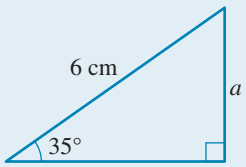
eles-4804

- When one acute angle and one side length are known in a right-angled triangle, this information can be used to find all other unknown sides or angles.

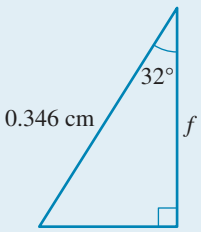
WORKED EXAMPLE 11 Using trigonometry to calculate side lengths

Calculate the value of each pronumeral, giving answers correct to 3 decimal places.

a.



b.

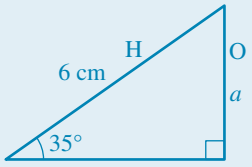


THINK

a. 1. Label the marked sides of the triangle.

WRITE/DRAW

a.



2. Identify the appropriate trigonometric ratio to use.

$$\sin(\theta) = \frac{O}{H}$$

3. Substitute $O = a$, $H = 6$ and $\theta = 35°$.

$$\sin(35°) = \frac{a}{6}$$

4. Make a the subject of the equation.

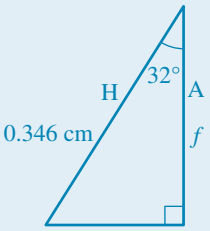
$$6\sin(35°) = a$$
$$x = 6\sin(35°)$$

5. Calculate and round the answer, correct to 3 decimal places.

$$a \approx 3.441 \text{ cm}$$

b. 1. Label the marked sides of the triangle.

b.



2. Identify the appropriate trigonometric ratio to use.

$$\cos(\theta) = \frac{\text{A}}{\text{H}}$$

3. Substitute A = f, H = 0.346 and $\theta = 32°$.

$$\cos(32°) = \frac{f}{0.346}$$

4. Make f the subject of the equation.

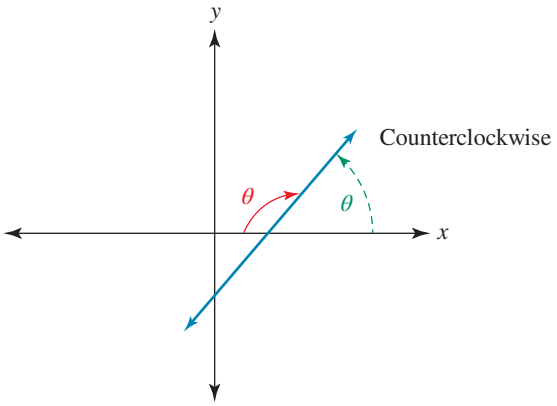
$$0.346\cos(32°) = f$$
$$f = 0.346\cos(32°)$$

5. Calculate and round the answer, correct to 3 decimal places.

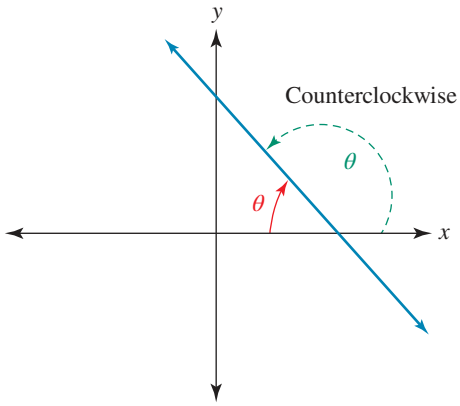
$$f \approx 0.293 \text{ cm}$$

5.5.2 Angle of inclination

- A **clinometer** is a tool used to measure angles of inclination. It is used in various fields, such as engineering, architecture and geology.
- An angle of inclination is the acute or obtuse angle that is formed when a non-horizontal line intersects the x-axis.
- It is measured counterclockwise from the x-axis to the line.
- In the figures shown, the angle in green is the angle of inclination since it is measured counterclockwise.



Positive slope

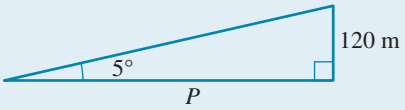


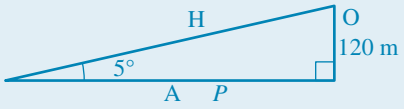
Negative slope

- When the slope is positive, the angle of inclination is an acute angle.
- When the slope is negative, the angle of inclination is an obtuse angle.

WORKED EXAMPLE 12 Using trigonometry to calculate side lengths

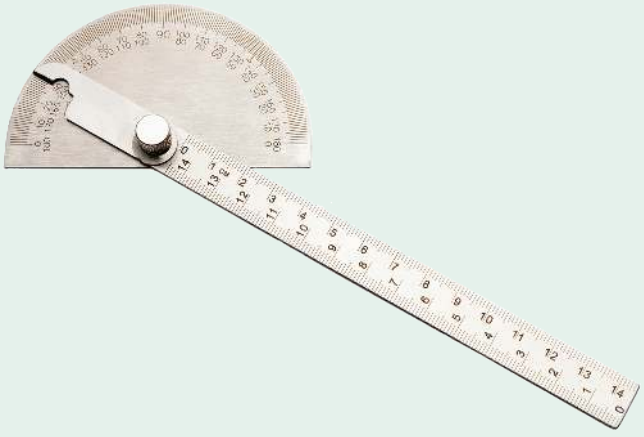
In the given triangle, the inclination angle is 5°. Calculate the value of the pronumeral *P*. Give the answer correct to 2 decimal places.



THINK	WRITE/DRAW
1. Label the marked sides of the triangle.	
2. Identify the appropriate trigonometric ratio to use.	$\tan(\theta) = \frac{O}{A}$
3. Substitute O = 120, A = *P* and $\theta = 5°$.	$\tan(5°) = \frac{120}{p}$
4. Make *P* the subject of the equation. i. Multiply both sides of the equation by *P*. ii. Divide both sides of the equation by tan (5°).	$P \times \tan(5°) = 120$ $P = \frac{120}{\tan(5°)}$
5. Calculate and round the answer, correct to 2 decimal places.	$P \approx 1371.61$ m

COLLABORATIVE TASK: Clinometer

The following group activity helps students learn how to use a clinometer to measure angles of inclination.



Materials used

- Clinometers (one per group of 3 or 4 students)
- Measuring tape or ruler
- Pencils and paper
- Protractor
- Worksheets or data recording sheets
- Objects with known angles of inclination (optional)
- Access to an outdoor area or a large indoor space (with high ceilings)

Instructions

1. Explain to the students how to use the clinometer and how it is a valuable tool for accurately measuring angles of inclination.
2. Demonstrate how a clinometer works, explaining its different components (e.g. sighting mirror, scale and sighting line) and how to read the measurements.
3. Emphasise the need to hold the clinometer level and steady to ensure accurate readings.
4. Divide the students into groups of 3 or 4 and distribute the materials.

5. Have each group choose a location outside to measure the angles of inclination. This could be a hill, a building, a tree, or any other object with an inclined surface.
6. Students should measure the height of the object using the measuring tape or ruler. They should also measure the distance from the object.
7. Measure the angle of inclination of the object from their location. They should record their measurements on paper, along with the height and distance of the object.
8. Once all groups have completed their measurements, have them compare their results and discuss any discrepancies. This can be a good opportunity to discuss sources of error in the measurements and ways to improve accuracy.
9. As a class, discuss the different types of angles that the students measured (angles of elevation or angles of depression) and their practical applications.
10. Finally, have each group present their findings and measurements to the class, along with any insights they gained from the activity.

This activity can help students develop their measurement skills, spatial reasoning, and collaboration skills. It can also help them understand the practical applications of measuring angles of inclination.

Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Interactivities Individual pathway interactivity: Using trigonometry to calculate side lengths (int-4588)
Using trigonometry to calculate side lengths (int-6133)

Exercise 5.5 Using trigonometry to calculate side lengths

learnon

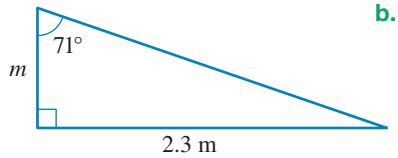
5.5 Quick quiz on | **5.5 Exercise**

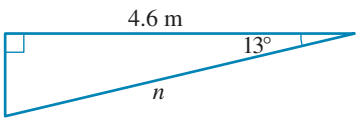
Individual pathways

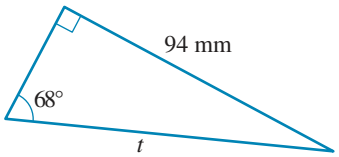
PRACTISE	CONSOLIDATE	MASTER
1, 2, 7, 9, 12	3, 5, 8, 10, 13	4, 6, 11, 14, 15

Fluency

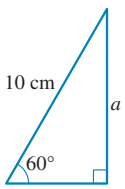
1. WE11 Calculate the value of the pronumeral in each of the following, correct to 2 decimal places.

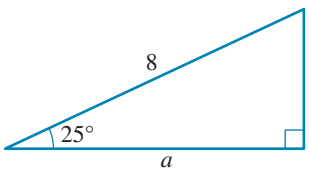
a.


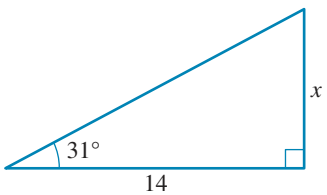
b.


c.


2. WE12 In each of the following triangles, the inclination angle is shown. Calculate the value of the pronumeral in each of the following, correct to 3 decimal places.

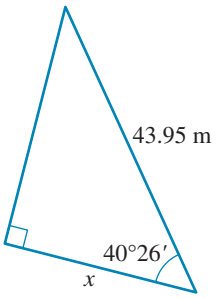
a.


b.


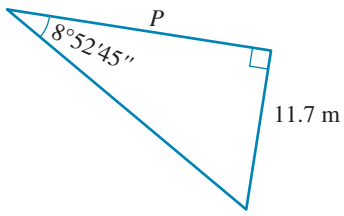
c.


3. Determine the length of the unknown side in each of the following, correct to 2 decimal places.

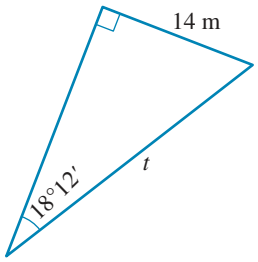
a.



b.

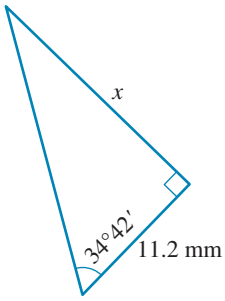


c.

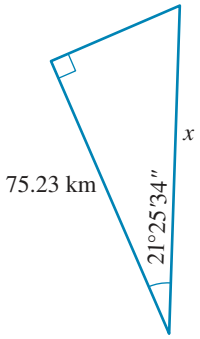


4. Determine the length of the unknown side in each of the following, correct to 2 decimal places.

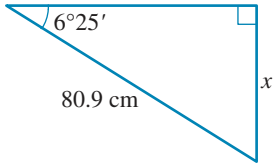
a.



b.

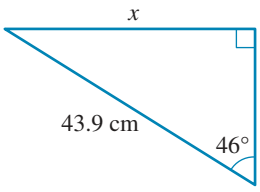


c.

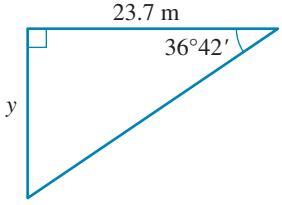


5. Calculate the value of the pronumeral in each of the following, correct to 2 decimal places.

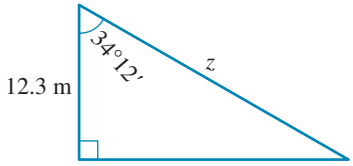
a.



b.

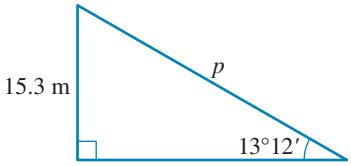


c.

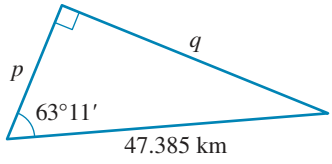


6. Calculate the value of the pronumeral in each of the following, correct to 2 decimal places.

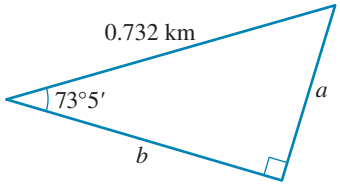
a.



b.



c.



Understanding

7. Given that the angle θ is 42° (the angle between the base and the hypotenuse of the right-angled triangle) and the length of the hypotenuse is 8.95 m in a right-angled triangle, calculate the length of:

a. the opposite side **b.** the adjacent side.

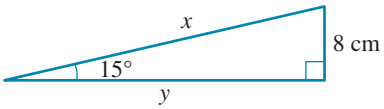
Give each answer correct to 1 decimal place.

8. A ladder rests against a wall. If the angle of inclination (between the ladder and the ground) is 35° and the foot of the ladder is 1.5 m from the wall, calculate how high up the wall the ladder reaches.
Write your answer in metres correct to 2 decimal places.

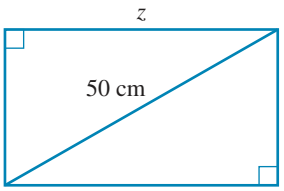
Reasoning

9. Tran is going to construct an enclosed rectangular desktop that has an inclination angle of 15°. The diagonal length of the desktop is 50 cm. At the high end, the desktop, including top, bottom and sides, will be raised 8 cm. The desktop will be made of wood.
The diagram below represents this information.

Side view of the desktop



Top view of the desktop



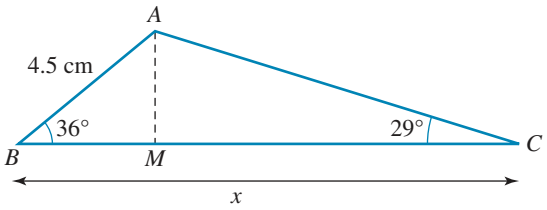
a. Determine the values (in centimetres) of x, y and z of the desktop. Write your answers correct to 2 decimal places.

b. Using your answer from part **a** determine the minimum area of wood, in cm^2, Tran needs to construct his desktop including top, bottom and sides. Write your answer correct to 2 decimal places.

10. a. In a right-angled triangle, under what circumstances will the opposite side and the adjacent side have the same length?

b. In a right-angled triangle, for what values of θ (the reference angle) will the adjacent side be longer than the opposite side?

11. In triangle ABC shown, the length of x correct to 2 decimal places is 8.41 cm. AM is perpendicular to BC. Jack found the length of x to be 5.11 cm. His working is shown below.

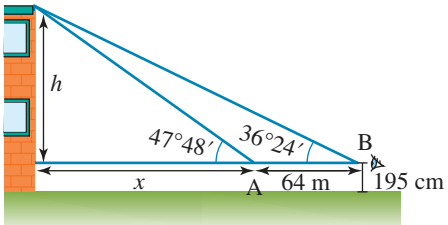


$$\cos(36°) = \frac{BM}{4.5}$$
$$BM = 3.641$$
$$\sin(36°) = \frac{AM}{4.5}$$
$$AM = 2.645$$
$$\tan(29°) = \frac{2.645}{MC}$$
$$MC = 1.466$$
$$BC = 3.641 + 1.466 = 5.107 \text{ cm}$$
$$= 5.11 \text{ cm (rounded to 2 decimal places)}$$

Identify his error and what he should have done instead.

Problem solving

12. A surveyor needs to determine the height of a building. She measures the angle to the top of the building from two points, A and B, which are 64 m apart. The surveyor's eye level is 195 cm above the ground.

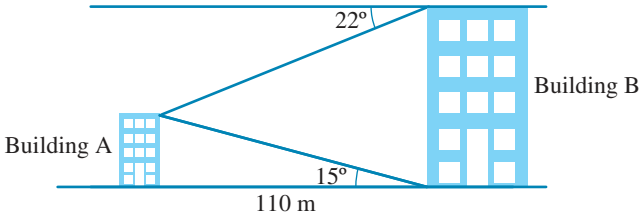


a. Determine the expressions for the height of the building, h, in terms of x using the two angles.

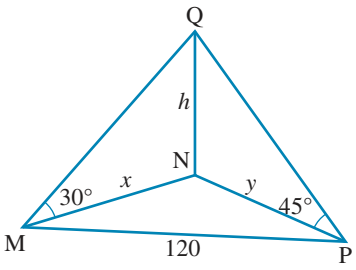
b. Solve for x by equating the two expressions obtained in part **a**. Give your answer to 2 decimal places.

c. Determine the height of the building correct to 2 decimal places.

13. Building A and Building B are 110 m apart. From the base of Building B to the top of Building A, the angle is 15°.
From the top of Building B looking down to the top of Building A, the angle is 22°.
Evaluate the height of each of the two buildings correct to 1 decimal place.



14. If angles QNM, QNP and MNP are right angles, determine the length of NQ.



15. Determine how solving a trigonometric equation differs when we are calculating the length of the hypotenuse side compared to when determining the length of a shorter side.

LESSON
5.6 Using trigonometry to calculate angle size

LEARNING INTENTION

At the end of this lesson you should be able to:
- apply inverse operations to calculate a known acute angle when two sides are given.

5.6.1 Using trigonometry to calculate angle size

eles-4805

- The size of any angle in a right-angled triangle can be found if the lengths of any two sides are known.
- Just as inverse operations are used to solve equations, inverse trigonometric ratios are used to solve trigonometric equations for the value of the angle.
  - Inverse sine ($\sin^{-1}$) is the inverse of sine.
  - Inverse cosine ($\cos^{-1}$) is the inverse of cosine.
  - Inverse tangent ($\tan^{-1}$) is the inverse of tangent.

Inverse operations

$$\text{If } \sin(\theta) = a, \text{ then } \sin^{-1}(a) = \theta.$$
$$\text{If } \cos(\theta) = a, \text{ then } \cos^{-1}(a) = \theta.$$
$$\text{If } \tan(\theta) = a, \text{ then } \tan^{-1}(a) = \theta.$$

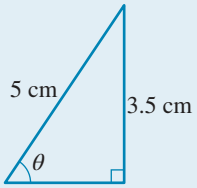
For example, since $\sin(30°) = 0.5$, then $\sin^{-1}(0.5) = 30°$; this is read as 'inverse sine of 0.5 is 30 degrees'.

A calculator can be used to calculate the values of inverse trigonometric ratios.

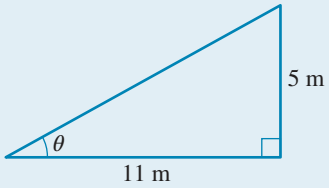
WORKED EXAMPLE 13 Evaluating angles using inverse trigonometric ratios

For each of the following, calculate the size of the angle, θ, correct to the nearest degree.

a.



b.

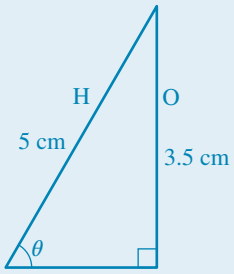


THINK

a. 1. Label the given sides of the triangle.

WRITE/DRAW

a.



2. Identify the appropriate trigonometric ratio to use. We are given O and H.

$$\sin(\theta) = \frac{O}{H}$$

3. Substitute O = 3.5 and H = 5 and evaluate the expression.

$$\sin(\theta) = \frac{3.5}{5}$$
$$= 0.7$$

4. Make θ the subject of the equation using inverse sine.

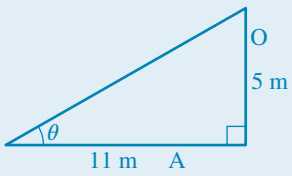
$$\theta = \sin^{-1}(0.7)$$
$$= 44.427\,004°$$

5. Evaluate θ and round the answer, correct to the nearest degree.

$$\theta \approx 44°$$

b. 1. Label the given sides of the triangle.

b.



2. Identify the appropriate trigonometric ratio to use. We are given O and A.

$$\tan(\theta) = \frac{O}{A}$$

3. Substitute O = 5 and A = 11.

$$\tan(\theta) = \frac{5}{11}$$

4. Make θ the subject of the equation using inverse tangent.

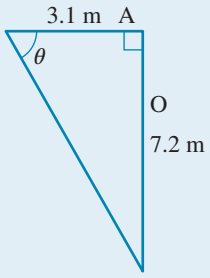
$$\theta = \tan^{-1}\left(\frac{5}{11}\right)$$
$$= 24.443\,954\,78°$$

5. Evaluate θ and round the answer, correct to the nearest degree.

$$\theta \approx 24°$$

WORKED EXAMPLE 14 Evaluating angles in minutes and seconds

Calculate the size of angle θ:
a. correct to the nearest second
b. correct to the nearest minute.

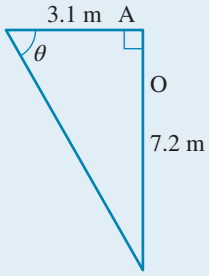


THINK

a. 1. Label the given sides of the triangle.

WRITE/DRAW

a.



2. Identify the appropriate trigonometric ratio to use.

$$\tan(\theta) = \frac{\text{O}}{\text{A}}$$

3. Substitute O = 7.2 and A = 3.1.

$$\tan(\theta) = \frac{7.2}{3.1}$$

4. Make θ the subject of the equation using inverse tangent.

$$\theta = \tan^{-1}\left(\frac{17.2}{3.1}\right)$$

5. Evaluate θ and write the calculator display.

$$\theta = 66.70543675°$$

6. Use the calculator to convert the answer to degrees, minutes and seconds.

$$\theta = 66°42'19.572''$$

7. Round the answer to the nearest second.

$$\theta \approx 66°42'20''$$

b. Round the answer to the nearest minute.

b. $\theta \approx 66°42'$

TI | THINK

a.

On a Calculator page, in degree mode, complete the entry line as:

$\tan^{-1}\left(\frac{7.2}{3.1}\right)$

To convert the decimal degree into degrees, minutes and seconds, press:

- CATALOG
- 1
- D

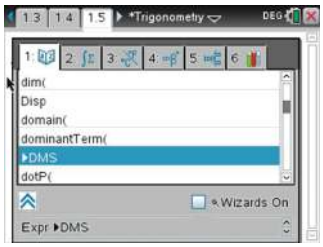
Scroll and select ▶ DMS. Then press ENTER.

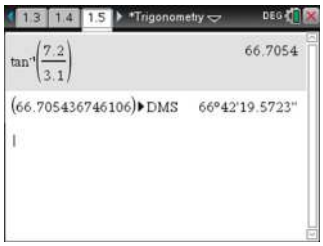
b.

Using the same screen, round to the nearest minute.

DISPLAY/WRITE

a.





$\theta = 66°42'20''$ correct to the nearest second.

b.

$\theta = 66°42'$ correct to the nearest minute.

CASIO | THINK

a.

On the Main screen in decimal and degree mode, complete the entry line as:

$\tan^{-1}\left(\frac{7.2}{3.1}\right)$

To convert the decimal answer into degrees, minutes and seconds, tap:

- Action
- Transformation
- DMS
- toDMS

Highlight the decimal answer and drag it into this line.

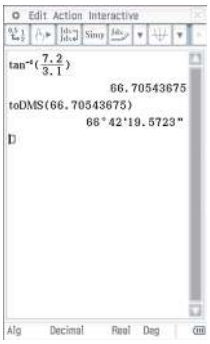
Then press EXE.

b.

Using the same screen, round to the nearest minute.

DISPLAY/WRITE

a.



$\theta = 66°42'20''$ correct to the nearest second.

b.

$\theta = 66°42'$ correct to the nearest minute.

Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Interactivities Individual pathway interactivity: Using trigonometry to calculate angle size (int-4589)
Finding the angle when two sides are known (int-6046)

Exercise 5.6 Using trigonometry to calculate angle size

learnon

5.6 Quick quiz on | **5.6 Exercise**

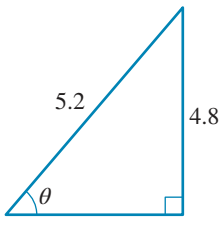
Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 7, 9, 12	2, 5, 8, 10, 13	3, 6, 11, 14

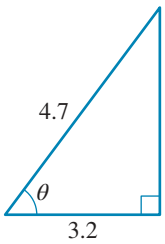
Fluency

1. WE13 Calculate the size of the angle, θ, in each of the following. Give your answers correct to the nearest degree.

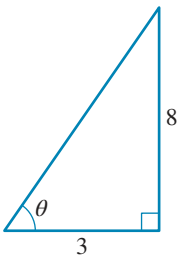
a.



b.

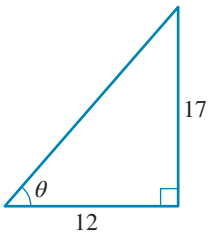


c.

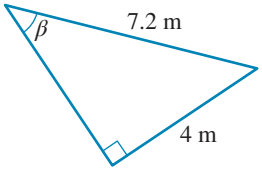


2. **WE14b** Calculate the size of the angle marked with the pronumeral in each of the following. Give your answers correct to the nearest minute.

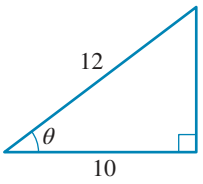
a.



b.

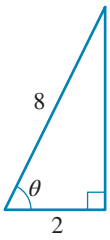


c.

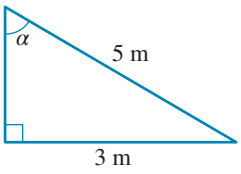


3. **WE14a** Calculate the size of the angle marked with the pronumeral in each of the following. Give your answers correct to the nearest second.

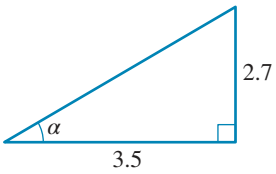
a.



b.

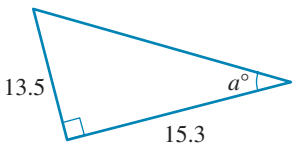


c.

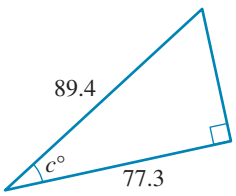


4. Calculate the size of the angle marked with the pronumeral in each of the following, giving your answers correct to the nearest degree.

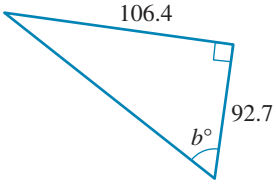
a.



b.

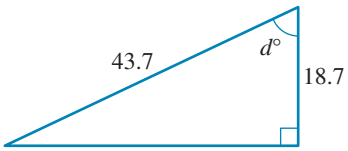


c.

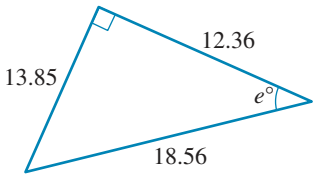


5. Calculate the size of the angle marked with the pronumeral in each of the following, giving your answers correct to the nearest degree.

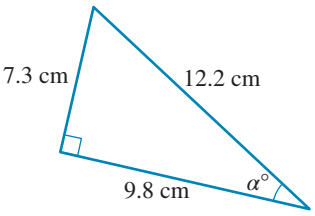
a.



b.

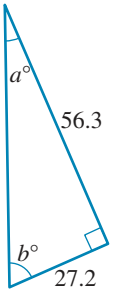


c.

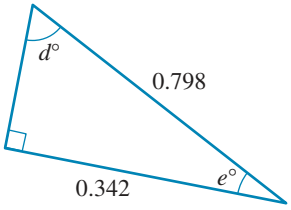


6. Calculate the size of each of the angles in the following, giving your answers correct to the nearest minute.

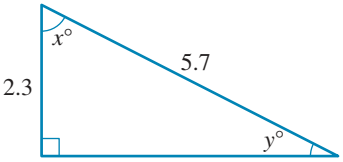
a.



b.

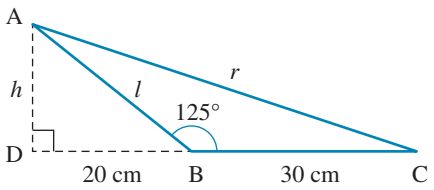


c.



Understanding

7. Answer the following questions for the triangle shown.



a. Calculate the length of the sides r, l and h. Write your answers correct to 2 decimal places.

b. Calculate the area of ABC, correct to the nearest square centimetre.

c. Determine the size of $\angle BCA$.

8. In the sport of air racing, small aeroplanes have to travel between two large towers (or pylons). The gap between a pair of pylons is smaller than the wingspan of the plane, so the plane has to go through on an angle with one wing higher than the other.
The wingspan of a competition plane is 8 metres.



a. Determine the angle, correct to 1 decimal place, that the plane has to tilt if the gap between pylons is:

i. 7 metres **ii.** 6 metres **iii.** 5 metres.

b. Because the plane has rolled away from the horizontal as it travels between the pylons it loses speed. If the plane's speed is below 96 km/h it will stall and possibly crash.
For each degree of 'tilt', the speed of the plane is reduced by 0.98 km/h. Calculate the minimum speed the plane must go through each of the pylons in part **a**.
Write your answer correct to 2 decimal places.

Reasoning

9. Explain how calculating the angle of a right-angled triangle is different to calculating a side length.

10. There are two important triangles commonly used in trigonometry.
Complete the following steps and answer the questions to create these triangles.

Triangle 1

- Sketch an equilateral triangle with side length 2 units.
- Calculate the size of the internal angles.
- Bisect the triangle to form two right-angled triangles.
- Redraw one of the triangles formed.
- Calculate the side lengths of this right-angled triangle as exact values.
- Fully label your diagram showing all side lengths and angles.

Triangle 2

- Draw a right-angled isosceles triangle.
- Calculate the sizes of the internal angles.
- Let the sides of equal length be 1 unit long each.
- Calculate the length of the third side as an exact value.
- Fully label your diagram showing all side lengths and angles.

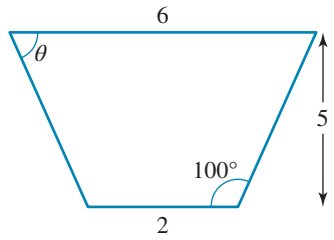
11. a. Use the triangles formed in question **10** to calculate exact values for sin(30°), cos(30°) and tan(30°). Justify your answers.

b. Use the exact values for sin(30°), cos(30°) and tan(30°) to show that $\tan(30°) = \dfrac{\sin(30°)}{\cos(30°)}$.

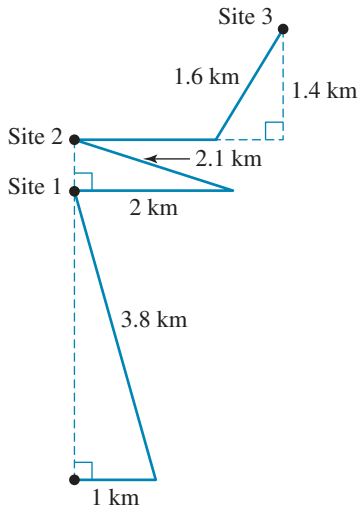
c. Use the formulas $\sin(\theta) = \dfrac{O}{H}$ and $\cos(\theta) = \dfrac{A}{H}$ to prove that $\tan(\theta) = \dfrac{\sin(\theta)}{\cos(\theta)}$.

Problem solving

12. Determine the angle θ in degrees and minutes.



13. During a Science excursion, a class visited an underground cave to observe rock formations. They were required to walk along a series of paths and steps as shown in the diagram below.



a. Calculate the angle of the incline (slope) required to travel down between each site. Give your answers to the nearest whole number.

b. Determine which path would have been the most challenging; that is, which path had the steepest slope.

14. At midday, the hour hand and the minute hand on a standard clock are both pointing at the twelve.



Calculate the angles the minute hand and the hour hand have moved 24.5 minutes later. Express both answers in degrees and minutes.

LESSON
5.7 Angles of elevation and depression

LEARNING INTENTION

At the end of this lesson you should be able to:

- identify angles of elevation and depression and solve for unknown side lengths and angles.

5.7.1 Angles of elevation and depression

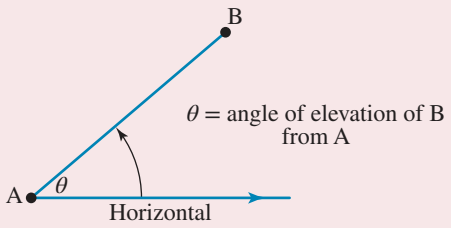
eles-4806

- Solving real-life problems usually involves the person measuring angles or lengths from their position using trigonometry.
- They may have to either look up at the object or look down to it; hence the terms '**angle of elevation**' and '**angle of depression**' respectively.

Angle of elevation

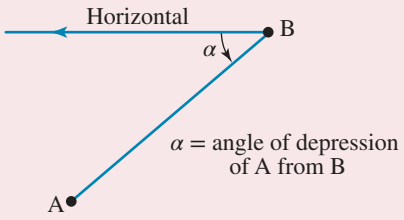
Consider the points A and B, where B is at a higher elevation than A.

If a horizontal line is drawn from A as shown, forming the angle θ, then θ is called the angle of elevation of B *from* A.



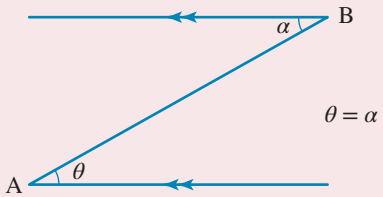
Angle of depression

If a horizontal line is drawn from B, forming the angle α, then α is called the angle of depression of A *from* B.



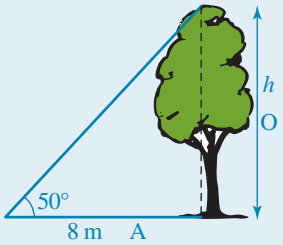
Alternate angle rule

Because the horizontal lines are parallel, θ and α have the same size (alternate angles).



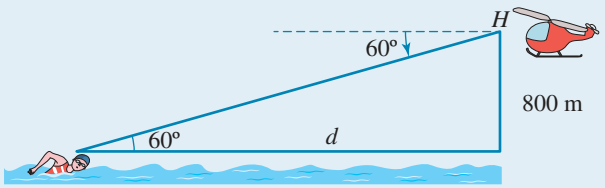
WORKED EXAMPLE 15 Applying angles of elevation to solve problems

From a point P, on the ground, the angle of elevation of the top of a tree is 50°. If P is 8 metres from the tree, determine the height of the tree correct to 2 decimal places.

THINK	WRITE/DRAW
1. Let the height of the tree be h. Sketch a diagram and show the relevant information.	
2. Identify the appropriate trigonometric ratio.	$\tan(\theta) = \frac{O}{A}$
3. Substitute $O = h$, $A = 8$ and $\theta = 50°$.	$\tan(50°) = \frac{h}{8}$
4. Rearrange to make h the subject.	$h = 8\tan(50°)$
5. Calculate and round the answer to 2 decimal places.	≈ 9.53
6. Write the answer in a sentence.	The height of the tree is 9.53 m.

WORKED EXAMPLE 16 Applying angles of depression to solve problems

The angle of depression from a helicopter, at point H, to a swimmer in distress in the water is 60°. If the helicopter is hovering 800 m above sea level, determine how far horizontally the swimmer is from the helicopter.
Write your answer in metres correct to 2 decimal places.

THINK	WRITE/DRAW
1. Let the horizontal distance between the swimmer and the helicopter be d. Sketch a diagram and show the relevant information.	
2. Identify the appropriate trigonometric ratio.	$\tan(\theta) = \frac{O}{A}$
3. Substitute $O = 800$, $Q = 60°$ and $A = d$.	$\tan(60°) = \frac{800}{d}$
4. Rearrange to make d the subject.	$d = \frac{800}{\tan(60°)}$
5. Calculate and round to 2 decimal places.	$d \approx 461.88$ m
6. Write the answer in a sentence.	The horizontal distance between the swimmer and the helicopter is 461.88 m.

Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

Interactivities Individual pathway interactivity: Angles of elevation and depression (int-4590)
Finding the angle of elevation and angle of depression (int-6047)

Exercise 5.7 Angles of elevation and depression

learnon

5.7 Quick quiz | 5.7 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 4, 6, 9, 12, 15	2, 5, 7, 10, 13, 16	3, 8, 11, 14, 17

Fluency

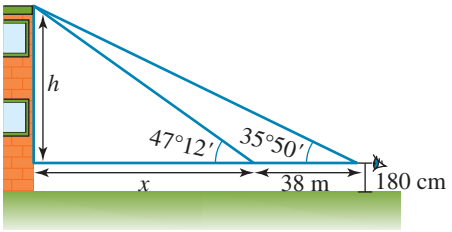
1. WE15 From a point P on the ground the angle of elevation from an observer to the top of a tree is 54°22′. If the tree is known to be 12.19 m high, determine how far P is from the tree (measured horizontally). Write your answer in metres correct to 2 decimal places.
2. WE16 From the top of a cliff 112 m high, the angle of depression to a boat is 9°15′. Determine how far the boat is from the foot of the cliff. Write your answer in metres correct to 1 decimal place.
3. A person on a ship observes a lighthouse on the cliff, which is 830 metres away from the ship. The angle of elevation of the top of the lighthouse is 12°.
   a. Determine how far above sea level the top of the lighthouse is, correct to 2 decimal places.
   b. If the height of the lighthouse is 24 m, calculate the height of the cliff, correct to 2 decimal places.



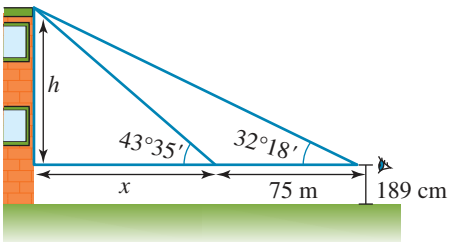
4. At a certain time of the day a post that is 4 m tall casts a shadow of 1.8 m. Calculate the angle of elevation of the sun at that time. Write your answer correct to the nearest minute.
5. An observer who is standing 47 m from a building measures the angle of elevation of the top of the building as 17°. If the observer's eye is 167 cm from the ground, determine the height of the building. Write your answer in metres correct to 2 decimal places.

Understanding

6. A surveyor needs to determine the height of a building. She measures the angle of elevation of the top of the building from two points that are 38 m apart. The surveyor's eye level is 180 cm above the ground.
   a. Determine two expressions for the height of the building, h, in terms of x using the two angles.
   b. Solve for x by equating the two expressions obtained in a. Write your answer in metres correct to 2 decimal places
   c. Determine the height of the building, in metres correct to 2 decimal places.



7. The height of another building needs to be determined but cannot be found directly. The surveyor decides to measure the angle of elevation of the top of the building from different sites, which are 75 m apart.
The surveyor's eye level is 189 cm above the ground.

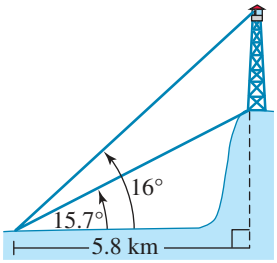


a. Determine two expressions for the height of the building above the surveyor's eye level, h, in terms of x using the two angles.

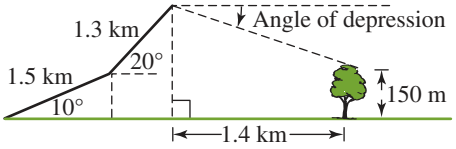
b. Solve for x. Write your answer in metres correct to 2 decimal places.

c. Determine the height of the building in metres correct to 2 decimal places.

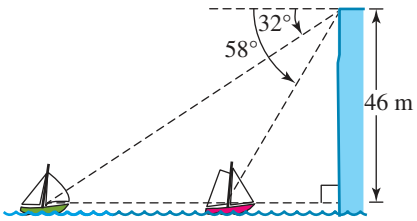
8. A lookout tower has been erected on top of a cliff. At a distance of 5.8 km from the foot of the cliff, the angle of elevation to the base of the tower is 15.7° and to the observation deck at the top of the tower is 16° respectively, as shown in the figure below.
Determine how high from the top of the cliff the observation deck is, to the nearest metre.



9. Elena and Sonja were on a camping trip to the Grampians, where they spent their first day hiking. They first walked 1.5 km along a path inclined at an angle of 10° to the horizontal.
Then they had to follow another path, which was at an angle of 20° to the horizontal. They walked along this path for 1.3 km, which brought them to the edge of the cliff. Here Elena spotted a large gum tree 1.4 km away.
If the gum tree is 150 m high, calculate the angle of depression from the top of the cliff to the top of the gum tree. Express your answer in degrees correct to the nearest degree.



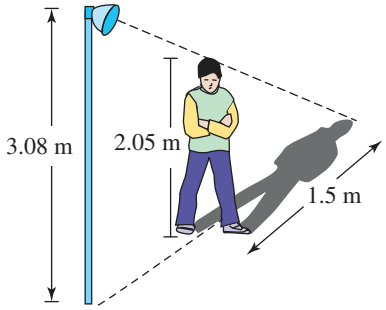
10. From a point on top of a cliff, two boats are observed. If the angles of depression are 58° and 32° and the cliff is 46 m above sea level, determine how far apart the boats are, in metres correct to 2 decimal places.



11. A 2.05-m tall man, standing in front of a street light 3.08 m high, casts a 1.5-m shadow.

a. Calculate the angle of elevation, to the nearest degree, from the ground to the source of light.

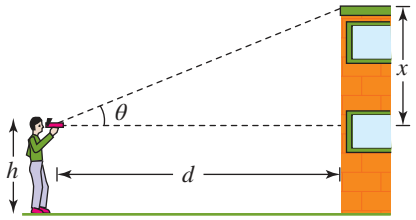
b. Determine how far the man is from the bottom of the light pole in metres correct to 2 decimal places.



Reasoning

12. Explain the difference between an angle of elevation and an angle of depression.

13. Joseph is asked to obtain an estimate of the height of his house using any mathematical technique. He decides to use a clinometer and basic trigonometry.
Using the clinometer, Joseph determines the angle of elevation, θ, from his eye level to the top of his house to be 42°.
The point from which Joseph measures the angle of elevation is 15 m away from his house and the distance from Joseph's eyes to the ground is 1.76 m.



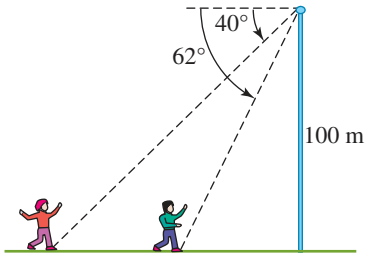
a. Determine the values for the pronumerals h, d and θ.
b. Determine the height of Joseph's house, in metres correct to 2 decimal places.

14. The angle of elevation of a vertically rising hot air balloon changes from 27° at 7:00 am to 61° at 7:03 am, according to an observer who is 300 m away from the take-off point.

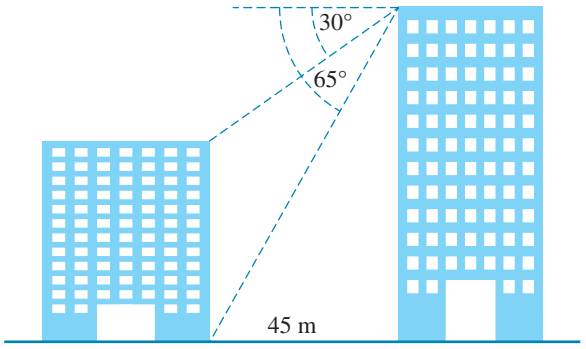
a. Assuming a constant speed, calculate that speed (in m/s and km/h) at which the balloon is rising, correct to 2 decimal places.
b. The balloon then falls 120 metres. Determine the angle of elevation now.
Write your answer in degrees correct to 1 decimal place.

Problem solving

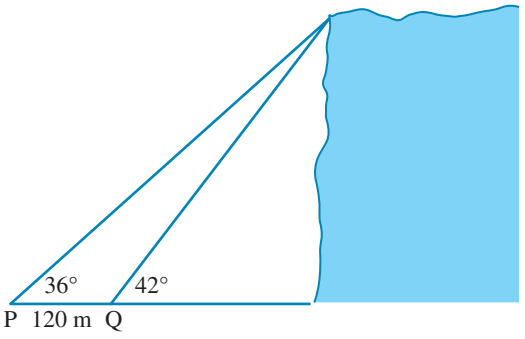
15. The competitors of a cross-country run are nearing the finish line. From a lookout 100 m above the track, the angles of depression to the two leaders, Nathan and Rachel, are 40° and 62° respectively.
Evaluate how far apart, to the nearest metre, the two competitors are.



16. The angle of depression from the top of one building to the foot of another building across the same street and 45 metres horizontally away is 65°.
The angle of depression to the roof of the same building is 30°. Evaluate the height of the shorter building.
Write your answer in metres correct to 3 decimal places.



17. P and Q are two points on a horizontal line that are 120 metres apart. The angles of elevation from P and Q to the top of a mountain are 36° and 42° respectively.
Determine the height of the mountain, in metres, correct to 1 decimal place.



LESSON
5.8 Bearings

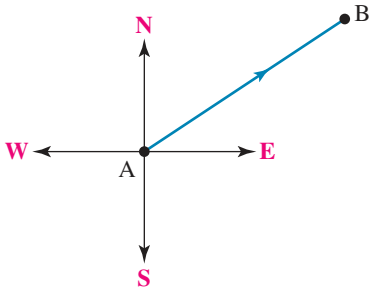
LEARNING INTENTIONS

At the end of this lesson you should be able to:
- draw diagrams with correct angles to represent information to help solve triangles
- apply trigonometry to solve bearing problems involving compass bearings and true bearings.

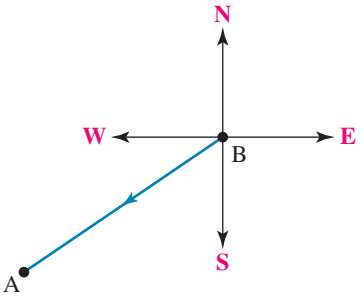
5.8.1 Using bearings

eles-4807

- A bearing gives the direction of travel from one point or object to another.
- The bearing of B from A tells how to get to B *from* A. A compass rose would be drawn at A.



- To illustrate the bearing of A *from* B, a compass rose would be drawn at B.



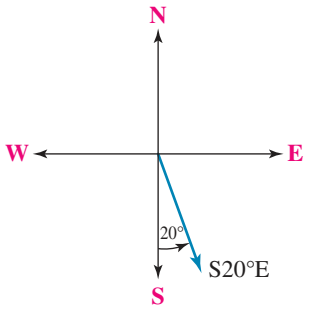
- There are two ways in which bearings are commonly written. They are compass bearings and true bearings.

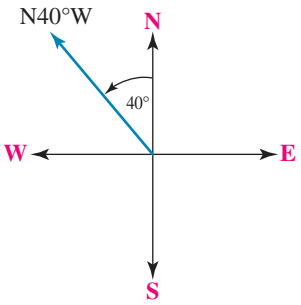
Compass bearings

- A **compass bearing** (for example N40°E or S72°W) has three parts.
  - The first part is either N or S (for north or south).
  - The second part is an acute angle.
  - The third part is either E or W (for east or west).

  For example, the compass bearing S20°E means start by facing south and then turn 20° towards the east. This is the direction of travel.

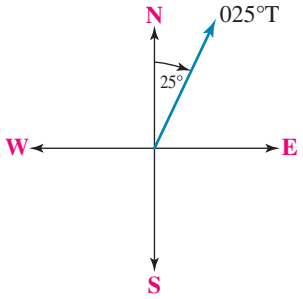
  N40°W means start by facing north and then turn 40° towards the west.

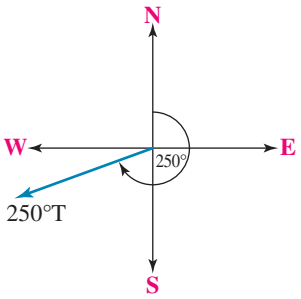




True bearings

- **True bearings** are measured from north in a clockwise direction and are expressed in 3 digits.
- The diagrams below show the bearings of 025° true and 250° true respectively. (These true bearings are more commonly written as 025°T and 250°T.)





WORKED EXAMPLE 17 Solving trigonometric problems involving bearings

A boat travels a distance of 5 km from P to Q in a direction of 035°T.

a. Calculate how far east of P is Q, correct to 2 decimal places.

b. Calculate how far north of P is Q, correct to 2 decimal places.

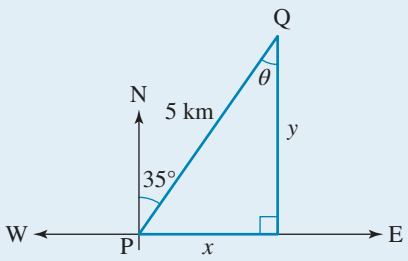
c. Calculate the true bearing of P from Q.

THINK

a. 1. Draw a diagram showing the distance and bearing of Q from P. Complete a right-angled triangle travelling x km due east from P and then y km due north to Q.

WRITE/DRAW

a.



2. To determine how far Q is east of P, we need to determine the value of x. We are given the length of the hypotenuse (H) and need to find the length of the opposite side (O). Write the sine ratio.

$$\sin(\theta) = \frac{O}{H}$$

3. Substitute $O = x$, $H = 5$ and $\theta = 35°$.

$$\sin(35°) = \frac{x}{5}$$

4. Make x the subject of the equation.

$$x = 5\sin(35°)$$

5. Evaluate and round the answer, correct to 2 decimal places. ≈ 2.87

6. Write the answer in a sentence. Point Q is 2.87 km east of P.

b. 1. To determine how far Q is north of P, we need to find the value of y. This can be done in several ways, namely: using the cosine ratio, the tangent ratio, or Pythagoras' theorem. Write the cosine ratio.

b. $\cos(\theta) = \frac{A}{H}$

2. Substitute $A = y$, $H = 5$ and $\theta = 35°$. $\cos(35°) = \frac{y}{5}$

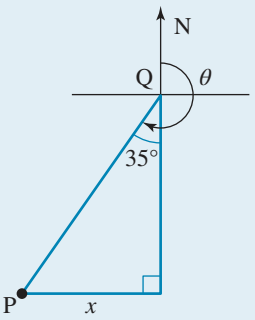
3. Make y the subject of the equation. $y = 5\cos(35°)$

4. Evaluate and round the answer, correct to 2 decimal places. ≈ 4.10

5. Write the answer in a sentence. Point B is 4.10 km north of A.

c. 1. To determine the bearing of P from Q, draw a compass rose at Q. The true bearing is given by $\angle\theta$.

c.



2. The value of θ is the sum of 180° (from north to south) and 35°. Write the value of θ.

$$\text{True bearing} = 180° + \alpha$$
$$\alpha = 35°$$
$$\text{True bearing} = 180° + 35°$$
$$= 215°$$

3. Write the answer in a sentence. The bearing of P from Q is 215°T.

5.8.2 Navigation problems involving bearings

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- Sometimes a journey includes a change in directions. In such cases, each section of the journey should be dealt with separately.

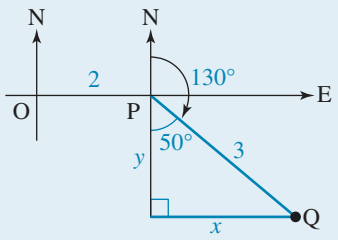
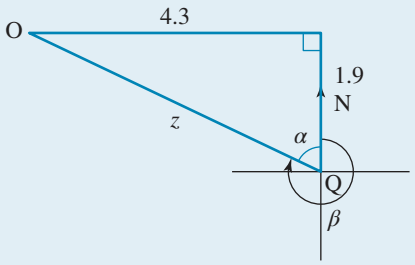
WORKED EXAMPLE 18 Solving bearings problems with two legs

A boy walks 2 km on a true bearing of 090° and then 3 km on a true bearing of 130°.

a. Calculate how far east of the starting point the boy is at the completion of his walk, correct to 1 decimal place.

b. Calculate how far south of the starting point the boy is at the completion of his walk, correct to 1 decimal place.

c. To return directly to his starting point, calculate how far the boy must walk and on what bearing. Write your answers in km correct to 2 decimal places and in degrees and minutes correct to the nearest minute.

THINK	WRITE/DRAW
a. 1. Draw a diagram of the boy's journey. The first leg of the journey is due east. Label the easterly component x and the southerly component y.	**a.** 
2. Write the ratio to determine the value of x.	$\sin(\theta) = \frac{O}{H}$
3. Substitute $O = x$, $H = 3$ and $\theta = 50°$.	$\sin(50°) = \frac{x}{3}$
4. Make x the subject of the equation.	$x = 3\sin(50°)$
5. Evaluate and round correct to 1 decimal place.	≈ 2.3 km
6. Add to this the 2 km east that was walked in the first leg of the journey and write the answer in a sentence.	Total distance east $= 2 + 2.3$ $= 4.3$ km The boy is 4.3 km east of the starting point.
b. 1. To determine the value of y (see the diagram in part **a**) we can use Pythagoras' theorem, as we know the lengths of two out of three sides in the right-angled triangle. Round the answer correct to 1 decimal place. *Note:* Alternatively, the cosine ratio could have been used.	**b.** Distance south $= y$ km $a^2 = c^2 - b^2$ $y^2 = 3^2 - 2.3^2$ $= 9 - 5.29$ $= 3.71$ $y = \sqrt{3.71}$ $= 1.9$ km
2. Write the answer in a sentence.	The boy is 1.9 km south of the starting point.
c. 1. Draw a diagram of the journey and write in the results found in parts **a** and **b**. Draw a compass rose at Q.	**c.** 
2. Determine the value of z using Pythagoras' theorem.	$z^2 = 1.9^2 + 4.3^2$ $= 22.1$ $z = 22.1$ ≈ 4.70
3. Determine the value of α using trigonometry.	$\tan(\alpha) = \frac{4.3}{1.9}$

4.	Make α the subject of the equation using the inverse tangent function.	$\alpha = \tan^{-1}\left(\frac{4.3}{1.9}\right)$
5.	Evaluate and round to the nearest minute.	$= 66.161259\,82°$ $= 66°9'40.535''$ $= 66°10'$
6.	The angle β gives the bearing.	$\beta = 360° - 66°10'$ $= 293°50'$
7.	Write the answer in a sentence.	The boy travels 4.70 km on a bearing of 293°50 T.

DISCUSSION

Explain the difference between true bearings and compass directions.

on Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)
Video eLesson Bearings (eles-1935)
Interactivities Individual pathway interactivity: Bearings (int-4591)
Bearings (int-6481)

Exercise 5.8 Bearings

learnon

5.8 Quick quiz on | 5.8 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 7, 10, 14, 17	2, 5, 8, 11, 12, 15, 18	4, 6, 9, 13, 16, 19

Fluency

1. Change each of the following compass bearings to true bearings.
a. N20°E **b.** N20°W **c.** S35°W

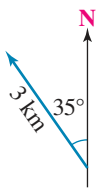
2. Change each of the following compass bearings to true bearings.
a. S28°E **b.** N34°E **c.** S42°W

3. Change each of the following true bearings to compass bearings.
a. 049°T **b.** 132°T **c.** 267°T

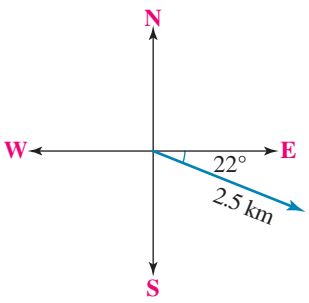
4. Change each of the following true bearings to compass bearings.
a. 330°T **b.** 086°T **c.** 234°T

5. Describe the following paths using true bearings.

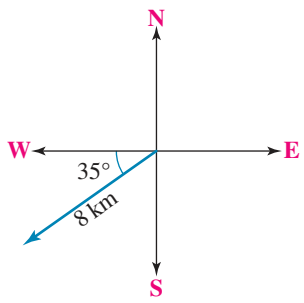
a.



b.

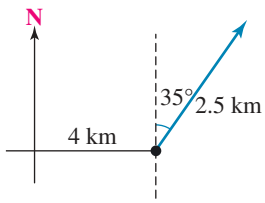


c.

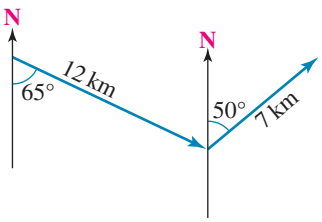


6. Describe the following paths using true bearings.

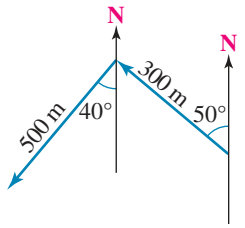
a.



b.



c.



7. Show each of the following journeys as a diagram.

a. A ship travels 040°T for 40 km and then 100°T for 30 km.

b. A plane flies for 230 km in a direction 135°T and a further 140 km in a direction 240°T.

8. Show each of the following journeys as a diagram.

a. A bushwalker travels in a direction 260°T for 0.8 km, then changes direction to 120°T for 1.3 km, and finally travels in a direction of 32° for 2.1 km.

b. A boat travels N40°W for 8 km, then changes direction to S30°W for 5 km and then S50°E for 7 km.

c. A plane travels N20°E for 320 km, N70°E for 180 km and S30°E for 220 km.

9. WE17 A yacht travels 20 km from *A* to *B* on a bearing of 042°T.



a. Calculate how far east of A is B, in km correct to 2 decimal places.

b. Calculate how far north of A is B, in km correct to 2 decimal places.

c. Calculate the bearing of A from B.

The yacht then sails 80 km from B to C on a bearing of 130°T.

d. Show the journey using a diagram.

e. Calculate how far south of B is C, in km correct to 2 decimal places.

f. Calculate how far east of B is C, in km correct to 2 decimal places.

g. Calculate the bearing of B from C.

10. If a farmhouse is situated 220 m N35°E from a shed, calculate the true bearing of the shed from the house.

Understanding

11. A pair of hikers travel 0.7 km on a true bearing of 240° and then 1.3 km on a true bearing of 300°. Calculate how far west have they travelled from their starting point, in km correct to 3 decimal places.

12. **WE18** A boat travels 6 km on a true bearing of 120° and then 4 km on a true bearing of 080°.

- **a.** Calculate how far east the boat is from the starting point on the completion of its journey, in km correct to 3 decimal places.
- **b.** Calculate how far south the boat is from the starting point on the completion of its journey, in km correct to 3 decimal places.
- **c.** Calculate the bearing of the boat from the starting point on the completion of its journey, correct to the nearest minute.

13. A plane flies on a true bearing of 320° for 450 km. It then flies on a true bearing of 350° for 130 km and finally on a true bearing of 050° for 330 km.
Calculate how far north of its starting point the plane is.
Write your answer in km correct to 2 decimal places.



Reasoning

14. A bushwalker leaves her tent and walks due east for 4.12 km, then walks a further 3.31 km on a bearing of N20°E. If she wishes to return directly to her tent, determine how far she must walk and what bearing she should take.
Write your answers in km correct to 2 decimal places and to the nearest degree.

15. A car travels due south for 3 km and then due east for 8 km. Determine the bearing of the car from its starting point, to the nearest degree. Show full working.

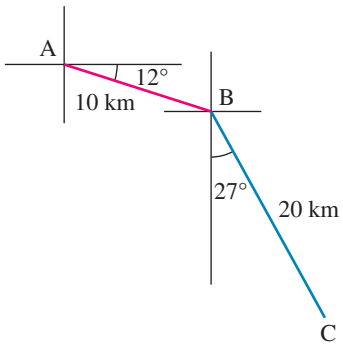
16. If the bearing of A from O is θ°T, then (in terms of θ) determine the bearing of O from A:

- **a.** if $0° < \theta° < 180°$
- **b.** if $180° < \theta° < 360°$.

Problem solving

17. A boat sails on a compass direction of E12°S for 10 km then changes direction to S27°E for another 20 km.
The pilot of the boat then decides to return to the starting point.

- **a.** Determine how far, correct to 2 decimal places, the boat is from its starting point.
- **b.** Determine on what bearing the boat should travel to return to its starting point. Write the angle correct to the nearest degree.



18. Samira and Tim set off early from the car park of a national park to hike for the day. Initially they walk N60°E for 12 km to see a spectacular waterfall.
They then change direction and walk in a south-easterly direction for 6 km, then stop for lunch.
Give all answers correct to 2 decimal places.

- **a.** Make a scale diagram of the hiking path they completed.
- **b.** Determine how far north of the car park they are at the lunch stop.
- **c.** Determine how far east of the car park they are at the lunch stop.
- **d.** Determine the bearing of the lunch stop from the car park.
- **e.** If Samira and Tim then walk directly back to the car park, calculate the distance they have covered after lunch.



19. Starting from their base in the national park, a group of bushwalkers travel 1.5 km at a true bearing of 030°, then 3.5 km at a true bearing of 160°, and then 6.25 km at a true bearing of 300°.
Evaluate how far, and at what true bearing, the group should walk to return to their base.
Write your answers in km correct to 2 decimal places and to the nearest degree.

LESSON
5.9 Applications

LEARNING INTENTIONS

At the end of this lesson you should be able to:
- draw well-labelled diagrams to represent information
- apply trigonometry to solve various problems involving triangles including bullet trajectory analysis..

5.9.1 Applications of trigonometry

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- When applying trigonometry to practical situations, it is essential to draw good mathematical diagrams using points, lines and angles.
- Several diagrams may be required to show all the necessary right-angled triangles.

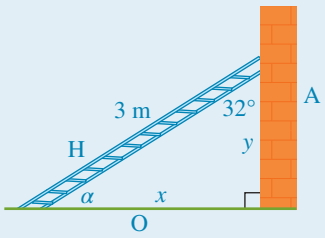
WORKED EXAMPLE 19 Applying trigonometry to solve problems

A ladder of length 3 m makes an angle of 32° with the wall.
a. Calculate how far the foot of the ladder is from the wall, in metres, correct to 2 decimal places.
b. Calculate how far up the wall the ladder reaches, in metres, correct to 2 decimal places.
c. Calculate the value of the angle the ladder makes with the ground.

THINK

Sketch a diagram and label the sides of the right-angled triangle with respect to the given angle.

WRITE/DRAW



a. 1. We need to calculate the distance of the foot of the ladder from the wall (O) and are given the length of the ladder (H). Write the sine ratio.

a. $\sin(\theta) = \frac{O}{H}$

2. Substitute $O = x$, $H = 3$ and $\theta = 32°$.

$\sin(32°) = \frac{x}{3}$

3. Make x the subject of the equation. $x = 3\sin(32°)$

4. Evaluate and round the answer to 2 decimal places. ≈ 1.59 m

5. Write the answer in a sentence. The foot of the ladder is 1.59 m from the wall.

b. 1. We need to calculate the height the ladder reaches up the wall (A) and are given the hypotenuse (H). Write the cosine ratio.

b. $\cos(\theta) = \frac{A}{H}$

2. Substitute $A = y$, $H = 3$ and $\theta = 32°$. $\cos(32°) = \frac{y}{3}$

3. Make y the subject of the equation. $y = 3\cos(32°)$

4. Evaluate and round the answer to 2 decimal places. $y \approx 2.54$ m

5. Write the answer in a sentence. The ladder reaches 2.54 m up the wall.

c. 1. To calculate the angle that the ladder makes with the ground, we could use any of the trigonometric ratios, as the lengths of all three sides are known. However, it is quicker to use the angle sum of a triangle.

c. $\alpha + 90° + 32° = 180°$
$\alpha + 122° = 180°$
$\alpha = 180° - 122°$
$\alpha = 58°$

2. Write the answer in a sentence. The ladder makes a 58° angle with the ground.

TI | THINK

On a Calculator page, complete the entry lines as:

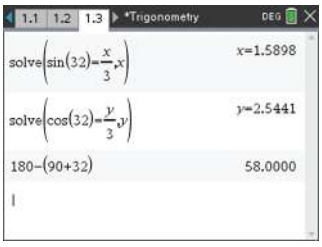
solve $\left(\sin(32) = \frac{x}{3}, x\right)$

solve $\left(\cos(32) = \frac{y}{3}, y\right)$

$180 - (90 + 32)$

Press ENTER after each entry.

DISPLAY/WRITE



$x = 1.59$ m correct to 2 decimal places.
$y = 2.54$ m correct to 2 decimal places.
$\alpha = 58°$

CASIO | THINK

On the Main screen, complete the entry lines as:

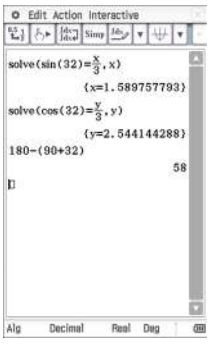
solve $\left(\sin(32) = \frac{x}{3}, x\right)$

solve $\left(\cos(32) = \frac{y}{3}, y\right)$

$180 - (90 + 32)$

Press EXE after each entry.

DISPLAY/WRITE



$x = 1.59$ m correct to 2 decimal places.
$y = 2.54$ m correct to 2 decimal places.
$\alpha = 58°$

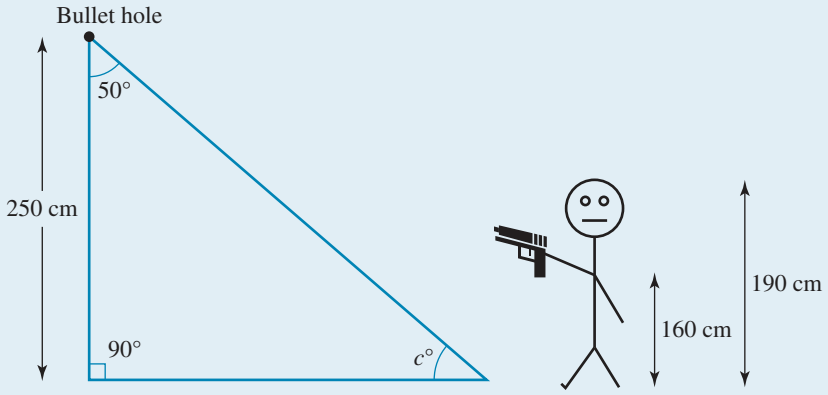
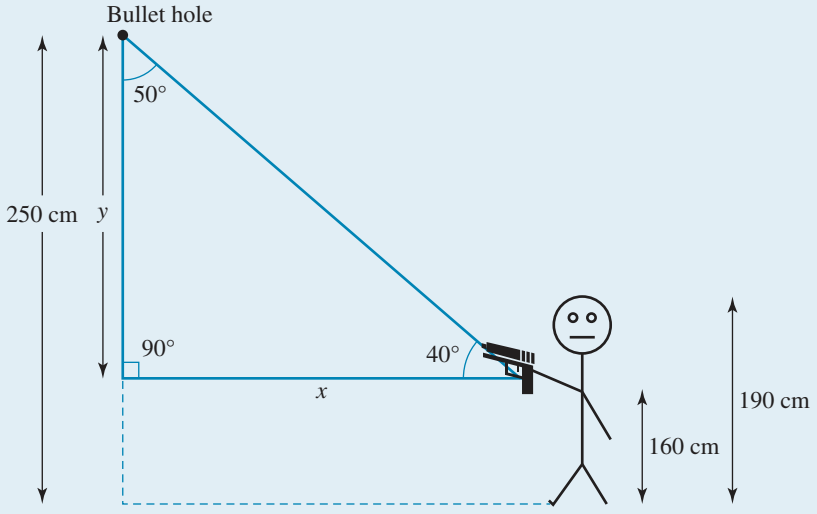
5.9.2 Bullet trajectory analysis

eles-6141

- Pythagoras' theorem and trigonometry can be used to model a crime scene.
- Trigonometry could be used to determine the trajectory of a bullet.
- Ballistics experts use these calculations to determine the impact angle and location of the shooter.

WORKED EXAMPLE 20 Calculating the location of a shooter

Police are investigating a shooting incident that resulted in a shot being fired that lodged in a wall. The height of the suspect is 190 cm, with a shoulder height of 160 cm. If the bullet hole is located at 250 cm from the floor and the bullet hole makes an angle of 50° with the wall (the impact angle), determine the distance of the shooter from the wall.

THINK	WRITE
1. Draw the diagram.	
2. As we know two angles in a right-angled triangle, we can calculate the value of the third angle.	Angle $c = 180° - 90° - 50°$ $= 40°$
3. Determine the length of the segment 'y'. This is the distance from the bullet hole to the shooter's shoulder height.	 $y = 250 - 160$ $= 90$ cm
4. Apply the tangent formula to determine the distance of the shooter from the wall.	$\tan 40° = \frac{90}{x}$ $x = \frac{90}{\tan 40°}$ $= 107.26$ cm
5. Write the answer.	The distance of the shooter from the wall is 107.26 cm.

COLLABORATIVE TASK: Modelling a crime scene

This activity introduces your students to a practical example of how a ballistics expert would gather evidence at a crime scene you can place 'bullet holes' around your classroom for your students to measure. This can be done with stickers on the wall or drawings on your whiteboard.

You might even want to cut some holes out of a foam core board for a 3D effect.

To help the students visualise the bullet trajectory you can use string or a laser pointer. The laser pointer is particularly effective when using the foam core bullet holes.

Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)
Interactivity Individual pathway interactivity: Applications (int-4592)

Exercise 5.9 Applications

learnon

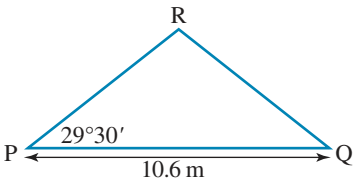
5.9 Quick quiz on | 5.9 Exercise

Individual pathways

PRACTISE	CONSOLIDATE	MASTER
1, 3, 4, 7, 12, 13, 16, 19	2, 5, 8, 10, 14, 17, 20	6, 9, 11, 15, 18, 21

Fluency

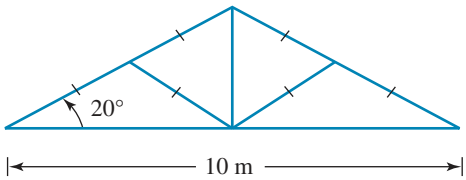
1. A carpenter wants to make a roof pitched at 29°30′, as shown in the diagram. Calculate how long, in metres correct to 2 decimal places, he should cut the beam PR.



2. The mast of a boat is 7.7 m high. A guy wire from the top of the mast is fixed to the deck 4 m from the base of the mast.
Determine the angle, correct to the nearest minute, that the wire makes with the horizontal.

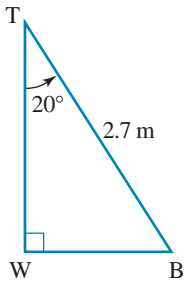
Understanding

3. A steel roof truss is to be made to the following design.
Write your answers in metres correct to 2 decimal places.

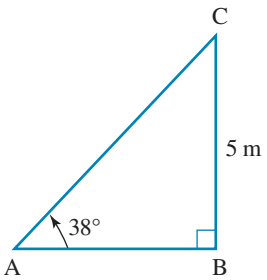


a. Calculate how high the truss is.
b. Determine the total length of steel required to make the truss.

4. **WE19** A ladder that is 2.7 m long is leaning against a wall at an angle of 20° as shown. If the base of the ladder is moved 50 cm further away from the wall, determine what angle the ladder will make with the wall.
Write your answer correct to the nearest minute.



5. A wooden framework is built as shown.
Bella plans to reinforce the framework by adding a strut from C to the midpoint of AB.
Calculate the length of the strut in metres correct to 2 decimal places.



6. Atlanta is standing due south of a 20 m flagpole at a point where the angle of elevation of the top of the pole is 35°. Ginger is standing due east of the flagpole at a point where the angle of elevation of the top of the pole is 27°.
Calculate how far, to the nearest metre, Ginger is from Atlanta.

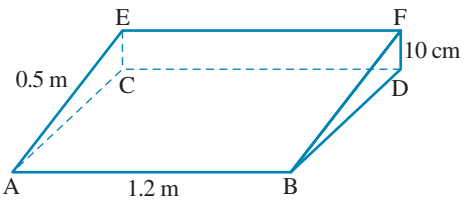
7. **WE20** Police are investigating a shooting incident that resulted in a shot being fired that lodged in a wall. The height of the suspect is 175 cm, with a shoulder height of 155 cm. If the bullet hole is located at 350 cm from the floor and the bullet hole makes an angle of 55° with the wall (the impact angle), determine the distance of the shooter from the wall.

8. A sniper is aiming at a target that is located on top of a building. The sniper is located on the ground, 500 meters away from the building. The height of the building is 40 meters. The sniper's rifle is elevated at an angle of 30 degrees above the horizontal.
Using Pythagoras' theorem and trigonometry, determine the height of the target above the ground.
Also assume that the bullet travels in a straight line and does not experience any air resistance.

9. From a point at ground level, Henry measures the angle of elevation of the top of a tall building to be 41°. After walking directly towards the building, he finds the angle of elevation to be 75°. If the building is 220 m tall, determine how far Henry walked between measurements.
Write your answer correct to the nearest metre.

10. Sailing in the direction of a mountain peak of height 893 m, Imogen measured the angle of elevation to be 14°. A short time later the angle of elevation was 27°.
Calculate how far, in km correct to 3 decimal places, Imogen had sailed in that time.

11. A desk top of length 1.2 m and width 0.5 m rises to 10 cm.

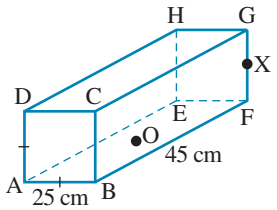


Calculate, correct to the nearest minute:

a. ∠DBF

b. ∠CBE.

12. A cuboid has a square end. If the length of the cuboid is 45 cm and its height and width are 25 cm each, calculate:

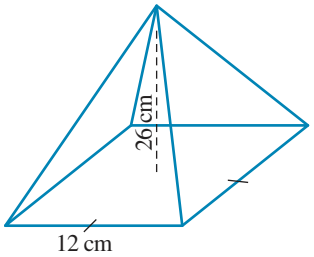


- **a.** the length of BD, correct to 2 decimal places
- **b.** the length of BG, correct to 2 decimal places
- **c.** the length of BE, correct to 2 decimal places
- **d.** the length of BH, correct to 2 decimal places
- **e.** ∠FBG, correct to the nearest minute
- **f.** ∠EBH, correct to the nearest minute.

If the midpoint of FG is X and the centre of the rectangle ABFE is O, calculate:

- **g.** the length OF, correct to 2 decimal places
- **h.** the length FX, correct to 1 decimal place
- **i.** ∠FOX, correct to the nearest minute
- **j.** the length OX, correct to 2 decimal places.

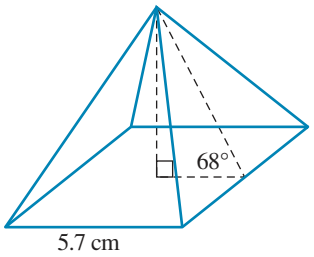
13. In a right square-based pyramid, the length of the side of the base is 12 cm and the height is 26 cm.



Determine:

- **a.** the angle the triangular face makes with the base, correct to the nearest degree
- **b.** the angle the sloping edge makes with the base, correct to the nearest minute
- **c.** the length of the sloping edge, in cm correct to 2 decimal places.

14. In a right square-based pyramid, the length of the side of the square base is 5.7 cm.



If the angle between the triangular face and the base is 68°, calculate:

- **a.** the height of the pyramid, in cm correct to 2 decimal places
- **b.** the angle the sloping edge makes with the base, correct to the nearest minute
- **c.** the length of the sloping edge, in cm correct to 2 decimal places.

15. In a right square-based pyramid, the height is 47 cm. If the angle between a triangular face and the base is 73°, calculate:

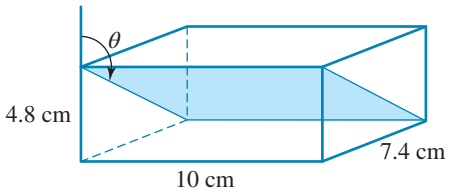
- **a.** the length of the side of the square base, in cm correct to 2 decimal places
- **b.** the length of the diagonal of the base, in cm correct to 2 decimal places
- **c.** the angle the sloping edge makes with the base, correct to the nearest minute.

Reasoning

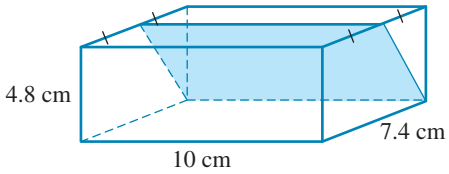
16. Explain whether sine of an acute angle can be 1 or greater.

17. A block of cheese is in the shape of a rectangular prism as shown. The cheese is to be sliced with a wide blade that can slice it in one go.
Calculate the angle (to the vertical, correct to 2 decimal places) that the blade must be inclined if:

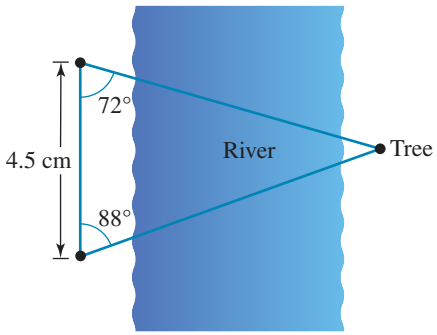
a. the block is to be sliced diagonally into two identical triangular wedges



b. the blade is to be placed in the middle of the block and sliced through to the bottom corner, as shown.

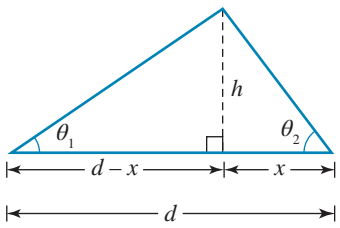


18. Aldo the carpenter is lost in a rainforest. He comes across a large river and he knows that he can not swim across it. Aldo intends to build a bridge across the river.
He draws some plans to calculate the distance across the river as shown in the diagram below.



a. Aldo used a scale of 1 cm to represent 20 m. Determine the real-life distance represented by 4.5 cm in Aldo's plans.

b. Use the diagram below to write an equation for h in terms of d and the two angles.



c. Use your equation from part **b** to find the distance across the river, correct to the nearest metre.

Problem solving

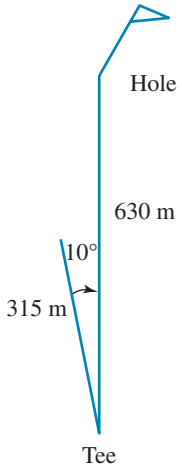
19. A ship travels north for 7 km, then on a true bearing of 140° for another 13 km.

a. Draw a sketch of the situation.

b. Determine how far south the ship is from its starting point, in km correct to 2 decimal places.

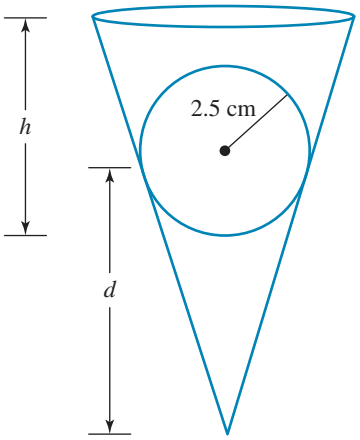
c. Evaluate the bearing, correct to the nearest degree, the ship is now from its starting point.

20. The ninth hole on a municipal golf course is 630 m from the tee. A golfer drives a ball from the tee a distance of 315 m at a 10° angle off the direct line as shown.



Determine how far the ball is from the hole and state the angle of the direct line that the ball must be hit along to go directly to the hole.
Give your answers correct to 1 decimal place.

21. A sphere of radius length 2.5 cm rests in a hollow inverted cone as shown. The height of the cone is 12.5 cm and its vertical angle is equal to 36°.



a. Evaluate the distance, d, from the tip of the cone to the point of contact with the sphere, correct to 2 decimal places.

b. Determine the distance, h, from the open end of the cone to the bottom of the ball, correct to 2 decimal places.

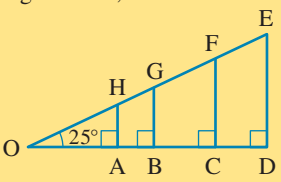
LESSON
5.10 Review

5.10.1 Topic summary

TRIGONOMETRY I

Similar triangles

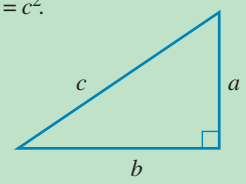
- When triangles have common angles, they are said to be similar.
  e.g. Triangles OAH, OBG and ODE are similar.



- Corresponding sides of similar triangles will have the same ratio.
  e.g. $\frac{FC}{OC} = \frac{ED}{OD} = \frac{HA}{OA}$

Pythagoras' theorem

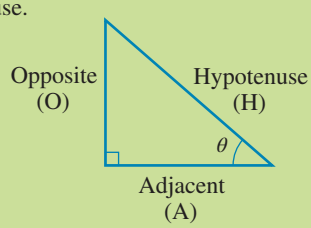
- When the length of two sides are known in a right-angled triangle, the third side can be found using the rule $a^2 + b^2 = c^2$.



- Length of the longest side $c = \sqrt{a^2 + b^2}$
- Length of the shorter sides $a = \sqrt{c^2 - b^2}$ or $b = \sqrt{c^2 - a^2}$

Trigonometric ratios (SOHCAHTOA)

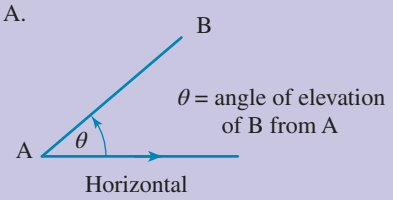
- In a right-angled triangle, the longest side is called the hypotenuse.



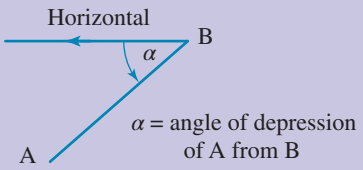
- If an acute angle is known, then the trigonometric ratios can be defined as:
  $$\sin\theta = \frac{O}{H}, \cos\theta = \frac{A}{H}, \tan\theta = \frac{O}{A}$$
- An acute angle can be calculated when two sides are known using the inverse operation of the correct trigonometric ratio.
  e.g. Since $\sin(30°) = 0.5$, then $\sin^{-1}(0.5) = 30°$; this is read as 'inverse sine of 0.5 is 30°'.

Angles of elevation and depression

- If a horizontal line is drawn from A as shown, forming the angle θ, then θ is called the **angle of elevation** of B *from* A.



- If a horizontal line is drawn from B as shown, forming the angle α, then α is called the **angle of depression** of A *from* B.

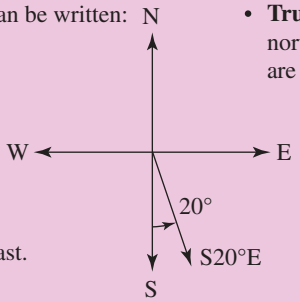


Bearings

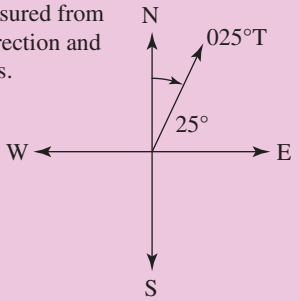
There are two ways in which bearings can be written:

- **Compass bearings** have 3 parts:
  - The first part is either N or S (for north or south).
  - The second part is an acute angle.
  - The third part is either E or W (for east or west).

  e.g. S20°E means start by facing south and then turn 20° towards the east.



- **True bearings** are measured from north in a clockwise direction and are expressed in 3 digits.



5.10.2 Success criteria

Tick the column to indicate that you have completed the lesson and how well you have understood it using the traffic light system.

(**Green:** I understand; **Yellow:** I can do it with help; **Red:** I do not understand)

Lesson	Success criteria			
5.2	I can identify similar right-angled triangles when corresponding sides are in the same ratio and corresponding angles are congruent.			
	I can apply Pythagoras' theorem to calculate the third side of a right-angled triangle when two other sides are known.			
5.3	I can apply Pythagoras' theorem to determine unknown lengths when a 3D diagram is given.			
	I can apply Pythagoras' theorem to determine unknown lengths in situations by first drawing a diagram.			
5.4	I can define trigonometric ratios according to the lengths of the relevant sides.			
	I can write equations for trigonometric ratios.			
5.5	I can apply trigonometric ratios to calculate the length of an unknown side when the length of one other side and an acute angle is known.			
5.6	I can apply inverse operations to calculate a known acute angle when two sides are given.			
5.7	I can identify angles of elevation and depression and solve for unknown side lengths and angles.			
5.8	I can draw diagrams with correct angles to represent information to help solve triangles.			
	I can apply trigonometry to solve bearing problems involving compass bearings and true bearings.			
5.9	I can draw well-labelled diagrams to represent information.			
	I can apply trigonometry to solve various problems involving triangles including bullet trajectory analysis.			

5.10.3 Project

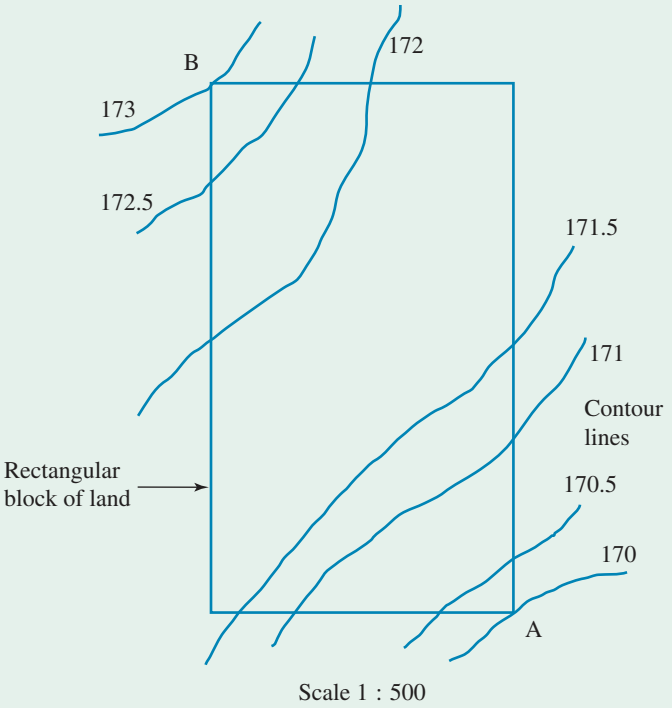
How steep is the land?

When buying a block of land on which to build a house, the slope of the land is often not very obvious. The slab of a house built on the ground must be level, so it is frequently necessary to remove or build up soil to obtain a flat area. The gradient of the land can be determined from a contour map of the area.



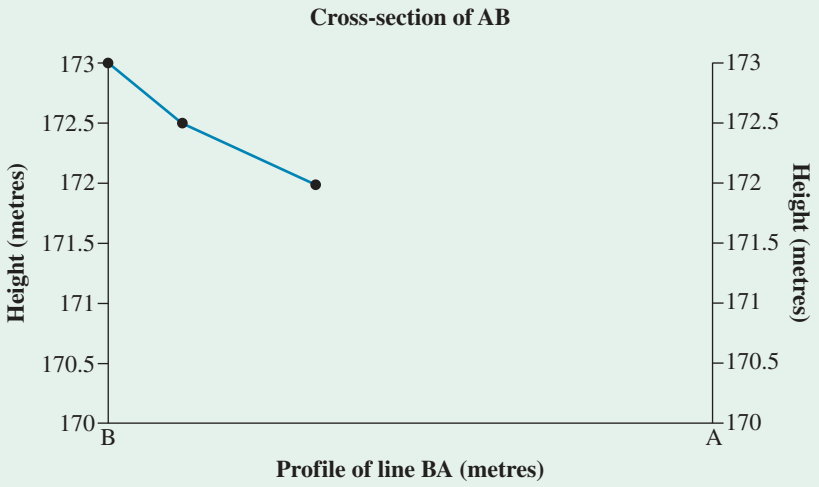
Consider the building block shown. The contour lines join points having the same height above sea level. Their measurements are in metres. The plan clearly shows that the land rises from A to B. The task is to determine the angle of this slope.

1. A cross-section shows a profile of the surface of the ground. Let us look at the cross-section of the ground between A and B. The technique used is as follows.
   - Place the edge of a piece of paper on the line joining A and B.
   - Mark the edge of the paper at the points where the contour lines intersect the paper.
   - Transfer this paper edge to the horizontal scale of the profile and mark these points.
   - Choose a vertical scale within the range of the heights of the contour lines.
   - Plot the height at each point where a contour line crosses the paper.
   - Join the points with a smooth curve.

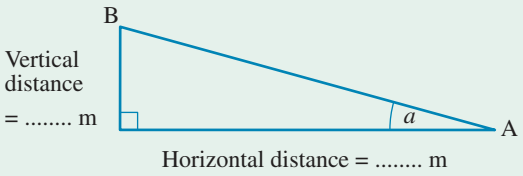


Scale 1 : 500

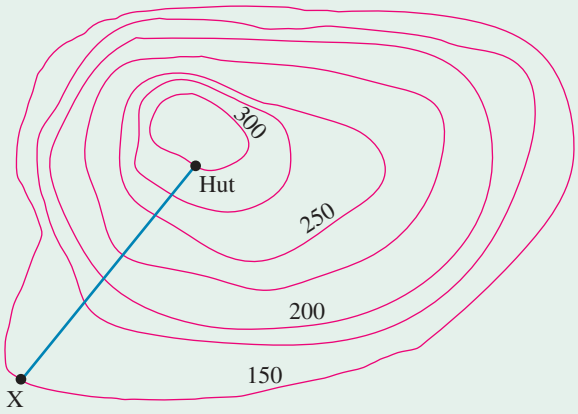
The cross-section has been started for you. Complete the profile of the line AB. You can now see a visual picture of the profile of the soil between A and B.



2. We now need to determine the horizontal distance between A and B.
   a. Measure the map distance between A and B using a ruler. What is the map length?
   b. Using the scale of 1 : 500, calculate the actual horizontal distance AB (in metres).
3. The vertical difference in height between A and B is indicated by the contour lines. Calculate this vertical distance.
4. Complete the measurements on this diagram.



5. The angle a represents the angle of the average slope of the land from A to B. Use the tangent ratio to calculate this angle (to the nearest minute).
6. In general terms, an angle less than 5° can be considered a gradual to moderate rise. An angle between 5° and 15° is regarded as moderate to steep while more than 15° is a steep rise. How would you describe this block of land?
7. Imagine that you are going on a bushwalk this weekend with a group of friends. A contour map of the area is shown. Starting at X, the plan is to walk directly to the hut.
   Draw a cross-section profile of the walk and calculate the average slope of the land. How would you describe the walk?



Scale 1 : 20 000

on Resources

eWorkbook Topic 5 Workbook (worksheets, code puzzle and project) (ewbk-2097)

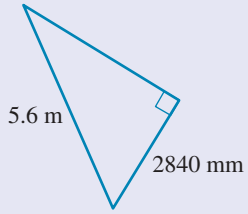
Interactivities Crossword (int-2869)
Sudoku puzzle (int-3592)

Exercise 5.10 Review questions

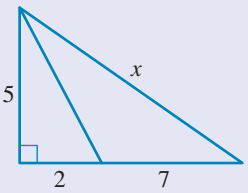
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Fluency

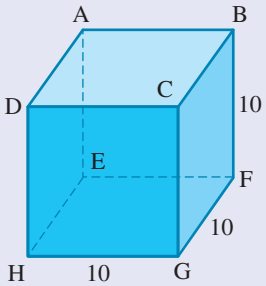
1. MC The most accurate measurement for the length of the third side in the triangle is:
   A. 483 m
   B. 23.3 cm
   C. 3.94 m
   D. 4826 mm



2. MC The value of x in this figure is:
   A. 5.4
   B. 7.5
   C. 10.1
   D. 10.3



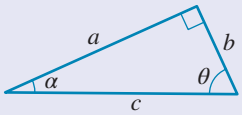
3. MC Select the closest length of AG of the cube.
   A. 10
   B. 30
   C. 20
   D. 17



4. MC If $\sin(38°) = 0.6157$, identify which of the following will also give this result.
   A. $\sin(218°)$
   B. $\sin(322°)$
   C. $\sin(578°)$
   D. $\sin(142°)$

5. MC The angle $118°52'34''$ is also equal to:
   A. $118.5234°$
   B. $118\frac{52}{34}°$
   C. $118.861°$
   D. $118.876°$

6. MC Identify which trigonometric ratio for the triangle shown below is incorrect.



A. $\sin(\alpha) = \frac{b}{c}$

B. $\sin(\alpha) = \frac{a}{c}$

C. $\cos(\alpha) = \frac{a}{c}$

D. $\tan(\alpha) = \frac{b}{a}$

7. MC Identify which of the following statements is correct.

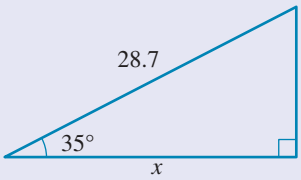
A. $\sin(55°) = \cos(55°)$

B. $\sin(45°) = \cos(35°)$

C. $\cos(15°) = \sin(85°)$

D. $\sin(42°) = \cos(48°)$

8. MC Identify which of the following can be used to determine the value of x in the diagram.



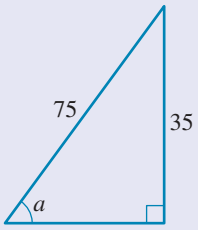
A. $28.7 \sin(35°)$

B. $28.7 \cos(35°)$

C. $28.7 \tan(35°)$

D. $\frac{28.7}{\sin(35°)}$

9. MC Identify which of the following expressions can be used to determine the value of a in the triangle shown.



A. $35 \sin(75°)$

B. $\sin^{-1}\left(\frac{35}{75}\right)$

C. $\sin^{-1}\left(\frac{75}{35}\right)$

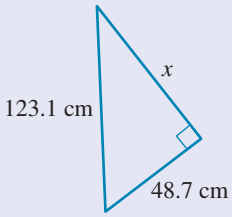
D. $\cos^{-1}\left(\frac{35}{75}\right)$

10. MC If a school is 320 m S42°W from the police station, calculate the true bearing of the police station from the school.

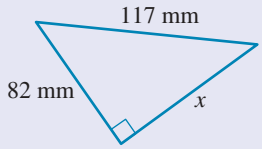
A. 042°T
B. 048°T
C. 222°T
D. 228°T

11. Calculate x, correct to 2 decimal places.

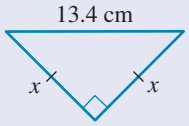
a.



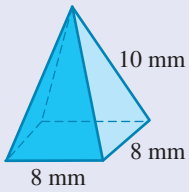
b.



12. Calculate the value of the pronumeral, correct to 2 decimal places.



13. Calculate the height of this pyramid, in mm correct to 2 decimal places.



14. A person standing 23 m away from a tree observes the top of the tree at an angle of elevation of 35°. If the person's eye level is 1.5 m from the ground, calculate the height of the tree, in metres correct to 1 decimal place.

15. A man with an eye level height of 1.8 m stands at the window of a tall building. He observes his young daughter in the playground below. If the angle of depression from the man to the girl is 47° and the floor on which the man stands is 27 m above the ground, determine how far from the bottom of the building the child is, in metres correct to 2 decimal places.

16. A plane flies 780 km in a direction of 185°T. Evaluate how far west it has travelled from the starting point, in km correct to 2 decimal places.

17. A hiker travels 3.2 km on a bearing of 250°T and then 1.8 km on a bearing of 320°T. Calculate how far west she has travelled from the starting point, in km correct to 2 decimal places.

18. If a 4 m ladder is placed against a wall and the foot of the ladder is 2.6 m from the wall, determine the angle (in degrees and minutes, correct to the nearest minute) the ladder makes with the wall.

Problem solving

19. The height of a right square-based pyramid is 13 cm. If the angle the face makes with the base is 67°, determine:

a. the length of the edge of the square base, in cm correct to 2 decimal places

b. the length of the diagonal of the base, in cm correct to 2 decimal places

c. the angle the slanted edge makes with the base in degrees and minutes, correct to the nearest minute.

20. A car is travelling northwards on an elevated expressway 6 m above ground at a speed of 72 km/h. At noon another car passes under the expressway, at ground level, travelling west, at a speed of 90 km/h.

a. Determine how far apart, in metres, the two cars are 40 seconds after noon, in metres correct to 2 decimal places.

b. At this time the first car stops, while the second car keeps going. Determine the time when they will be 3.5 km apart. Write your answer correct to the nearest tenth of a second.

21. Two towers face each other separated by a distance, d, of 20 metres. As seen from the top of the first tower, the angle of depression of the second tower's base is 59° and that of the top is 31°. Calculate the height, in metres correct to 2 decimal places, of each of the towers.

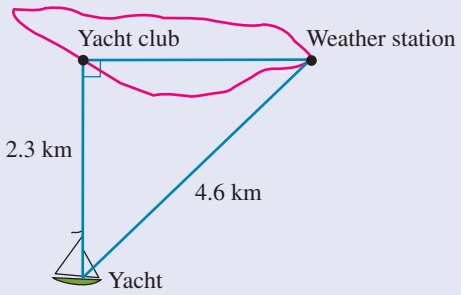
22. A piece of flat pastry is cut in the shape of a right-angled triangle. The longest side is $6b$ cm and the shortest is $2b$ cm.

a. Determine the length of the third side. Give your answer in exact form.

b. Determine the sizes of the angles in the triangle.

c. Show that the area of the triangle is equal to $4\sqrt{2}b^2$ cm^2.

23. A yacht is anchored off an island. It is 2.3 km from the yacht club and 4.6 km from a weather station. The three points form a right-angled triangle at the yacht club.



a. Calculate the angle at the yacht between the yacht club and the weather station.

b. Evaluate the distance between the yacht club and the weather station, in km correct to 2 decimal places.

The next day the yacht travels directly towards the yacht club but is prevented from reaching the club because of dense fog. The weather station notifies the yacht that it is now 4.2 km from the station.

c. Calculate the new angle at the yacht between the yacht club and the weather station, in degrees correct to 1 decimal place.

d. Determine how far the yacht is now from the yacht club, correct to 2 decimal places.



To test your understanding and knowledge of this topic, go to your learnON title at www.jacplus.com.au and complete the **post-test**.

Answers

Topic 5 Trigonometry I

5.1 Pre-test

1. $w = 6.89$ cm
2. $x = 2.24$ cm
3. 16.97 cm
4. 62.28 cm
5. 0.3521
6. $\theta = 55°10'$
7. $\theta = 36°52'12''$
8. $y = 6.7$ m
9. C
10. a. 020°T b. 227°T c. 327°T d. 163°T
11. A
12. D
13. 551.2 m
14. D
15. $x = 9.0$

5.2 Pythagoras' theorem

1. a. 7.86 b. 33.27 c. 980.95
2. a. 12.68 b. 2.85 c. 175.14
3. a. 36.36 b. 1.62 c. 15.37
4. a. 0.61 b. 2133.19 c. 453.90
5. 23.04 cm
6. 12.65 cm
7. a. 14.14 cm b. 24.04 cm c. 4.53 cm
8. a. 74.83 cm b. 249.67 cm c. 3741.66 cm^2
9. a. 6.06 b. 4.24 c. 4.74
10. 14.84 cm
11. 15.59 cm
12. 19.23 cm
13. 72.75 cm; 3055.34 cm^2
14. 39 m
15. 4.34 km
16. 38.2 m
17. 63.06 mm
18. a. 32 cm b. 768 cm^2
19. 26.83 diagonals, so would need to complete 27
20. 4701.06 m
21. 9.90 cm
22. a. 65 b. 185 c. 305
23. The value found using Pythagoras' theorem represents length and therefore can't be negative.
24. a. Neither 105 nor 208 can be the hypotenuse of the triangle, because they are the two smallest values. The other two values could be the hypotenuse if they enable the creation of a right-angled triangle.
    b. 105, 208, 233
25. a. 21 cm
    b. 35 cm
    c. $y = 12.6$ cm and RS = 9.8 cm
26. 13.86 cm
27. 1.33 million km

5.3 Pythagoras' theorem in three dimensions (10A)

1. a. 13.86 b. 13.93 c. 18.03
2. 12.21, 12.85
3. 4.84 m, 1.77 m
4. 8.49, 4.24
5. 31.62 cm
6. 10.58 cm
7. 23 mm
8. a. i. 233.24 m ii. 200.12 m iii. 120.20 m
   b. 116.83 m
9. 14.72 cm
10. 12.67 cm
11. 42.27 cm
12. Sample responses can be found in the worked solutions in the online resources.
13. 186.5 m
14. a. 1.49 m b. 7.43m^2
15. $\sqrt{1265}$ cm
16. 25.475
17. 28.6 m

5.4 Trigonometric ratios

1. a. 0.5000 b. 0.7071 c. 0.4663
   d. 0.8387 e. 8.1443 f. 0.7193
2. a. 0.6494 b. 0.5885 c. 0.5220
   d. −1.5013 e. 0.9990 f. 0.6709
3. a. 0.8120 b. 0.5253 c. −0.8031
   d. 0.4063 e. 0.9880
4. a. −0.9613 b. 1.7321 c. −0.5736 d. 0.1320
5. a. 50° b. 24° c. 53°
6. a. 71° b. 86° c. 41°
7. a. 54°29′ b. 6°19′ c. 0°52′
8. a. 72°47′ b. 44°48′ c. 26°45′
9. a. 26°33′54″ b. 64°1′25″ c. 64°46′59″
10. a. 48°5′22″ b. 36°52′12″ c. 88°41′27″
11. a. 2.824 b. 71.014 c. 20.361 d. 2.828
12. a. 226.735 b. 1.192 c. 7.232 d. 32.259
13. a. 4909.913 b. 0.063 c. 0.904 d. 14.814
14. a. $\sin(\theta) = \frac{e}{f}$ b. $\cos(\theta) = \frac{d}{f}$ c. $\tan(\theta) = \frac{e}{d}$
15. a. $\sin(\alpha) = \frac{i}{g}$ b. $\cos(\alpha) = \frac{h}{g}$ c. $\tan(\alpha) = \frac{i}{h}$

16. a. $\sin(\beta) = \frac{1}{k}$ b. $\cos(\beta) = \frac{j}{k}$ c. $\tan(\beta) = \frac{1}{j}$

17. a. $\sin(\gamma) = \frac{n}{\text{m}}$ b. $\cos(\gamma) = \frac{o}{\text{m}}$ c. $\tan(\gamma) = \frac{n}{o}$

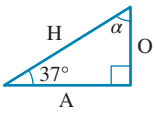
18. a. $\sin(\beta) = \frac{b}{c}$ b. $\cos(\beta) = \frac{a}{c}$ c. $\tan(\beta) = \frac{b}{a}$

19. a. $\sin(\gamma) = \frac{v}{u}$ b. $\cos(\gamma) = \frac{t}{u}$ c. $\tan(\gamma) = \frac{v}{t}$

20. a. $\sin(\theta) = \frac{15}{18}$ b. $\cos(\theta) = \frac{22}{30}$ c. $\tan(\theta) = \frac{7}{9}$

21. a. $\tan(\theta) = \frac{3.6}{p}$ b. $\sin(25°) = \frac{13}{t}$ c. $\sin(\alpha) = \frac{18.6}{23.5}$

22. a.



b. i. $\sin(37°) = 0.60$ ii. $\cos(37°) = 0.80$
iii. $\tan(37°) = 0.75$

c. $\alpha = 53°$

23. a. i. $\sin(53°) = 0.80$ ii. $\cos(53°) = 0.60$
iii. $\tan(53°) = 1.33$

b. They are equal.

c. They are equal.

d. The sine of an angle is equal to the cosine of its complement angle.

24. $\sin(\theta) = \frac{\text{opp}}{\text{hyp}}, \cos(\theta) = \frac{\text{adj}}{\text{hyp}} \Rightarrow \frac{\sin(\theta)}{\cos(\theta)} = \frac{\text{opp}}{\text{adj}} = \tan(\theta)$

25. a. $h^2 = a^2 - x^2$

b. $h^2 = c^2 - b^2 + 2bx - x^2$

c. Sample responses can be found in the worked solutions in the online resources.

d. Sample responses can be found in the worked solutions in the online resources.

26. $\text{DC} = x + \frac{y}{\tan(\theta)}$

27. To determine which trigonometric ratio to apply, the sides in relation to the angle relevant in the question need to be identified and named.

28. $110\tan(\theta + \alpha) - 110\tan(\alpha)$

5.5 Using trigonometry to calculate side lengths

1. a. 0.79 b. 4.72 c. 101.38

2. a. 8.660 b. 7.250 c. 8.412

3. a. 33.45 m b. 74.89 m c. 44.82 m

4. a. 7.76 mm b. 80.82 km c. 9.04 cm

5. a. $x = 31.58$ cm b. $y = 17.67$ m c. $z = 14.87$ m

6. a. $p = 67.00$ m

b. $p = 21.38$ km, $q = 42.29$ km

c. $a = 0.70$ km, $b = 0.21$ km

7. a. 6.0 m b. 6.7 m

8. 1.05 m

9. a. $x = 30.91$ cm, $y = 29.86$ cm, $z = 39.30$ cm

b. 2941.54 cm^2

10. a. In an isosceles right-angled triangle

b. $\theta < 45°$

11. Sample responses can be found in the worked solutions in the online resources.

12. a. $h = \tan(47°48')x$ m
$h = \tan(36°24')(x + 64)$ m

b. 129.07 m

c. 144.29 m

13. Building A is 29.5 m and Building B is 73.9 m.

14. 60

15. Sample responses can be found in the worked solutions in the online resources.

5.6 Using trigonometry to calculate angle size

1. a. 67° b. 47° c. 69°

2. a. 54°47′ b. 33°45′ c. 33°33′

3. a. 75°31′21″ b. 36°52′12″ c. 37°38′51″

4. a. 41° b. 30° c. 49°

5. a. 65° b. 48° c. 37°

6. a. $a = 25°47'$, $b = 64°13'$ b. $d = 25°23'$, $e = 64°37'$
c. $x = 66°12'$, $y = 23°48'$

7. a. $r = 57.58$, $l = 34.87$, $h = 28.56$

b. 428 cm^2

c. 29.7°

8. a. i. 29.0° ii. 41.4° iii. 51.3°

b. i. 124.42 km/h ii. 136.57 km/h iii. 146.27 km/h

9. To determine the size of acute angles, use inverse operations.

10. Sample responses can be found in the worked solutions in the online resources.

11. a. $\sin(30°) = \frac{1}{2}, \cos(30°) = \frac{\sqrt{3}}{2}, \tan(30°) = \frac{\sqrt{3}}{3}$

b. Sample responses can be found in the worked solutions in the online resources.

c. Sample responses can be found in the worked solutions in the online resources.

12. 58°3′

13. a. Between site 3 and site 2: 61°
Between site 2 and site 1: 18°
Between site 1 and bottom: 75°

b. Between site 1 and bottom: 75° slope

14. 147°0′; 12°15′

5.7 Angles of elevation and depression

1. 8.74 m

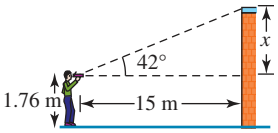
2. 687.7 m

3. a. 176.42 m b. 152.42 m

4. 65°46′

5. 16.04 m

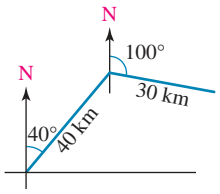
6. a. $h = x\tan(47°12')$ m; $h = (x + 38)\tan(35°50')$ m
   b. $x = 76.69$ m
   c. 84.62 m
7. a. $h = x\tan(43°35')$ m; $h = (x + 75)\tan(32°18')$ m
   b. 148.37 m
   c. 143.10 m
8. 0.033 km or 33 m
9. 22°
10. 44.88 m
11. a. 54° b. 0.75 m
12. Angle of elevation is an angle measured upwards from the horizontal. Angle of depression is measured from the horizontal downwards.
13. a.



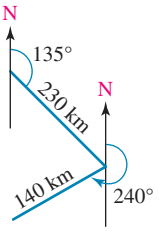
   b. 15.27 m
14. a. 2.16 m/s, 7.77 km/h b. 54.5°
15. 66 m
16. 70.522 m
17. 451.5 m

5.8 Bearings

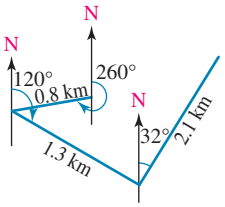
1. a. 020°T b. 340°T c. 215°T
2. a. 152°T b. 034°T c. 222°T
3. a. N49°E b. S48°E c. S87°W
4. a. N30°W b. N86°E c. S54°W
5. a. 3 km 325°T b. 2.5 km 112°T c. 8 km 235°T
6. a. 4 km 090°T, then 2.5 km 035°T
   b. 12 km 115°T, then 7 km 050°T
   c. 300 m 310°T, then 500 m 220°T
7. a.



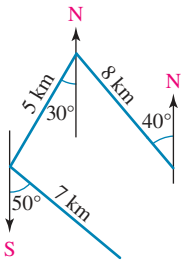
   b.



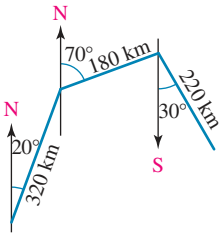
8. a.



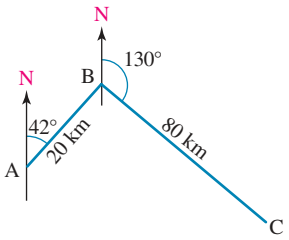
   b.



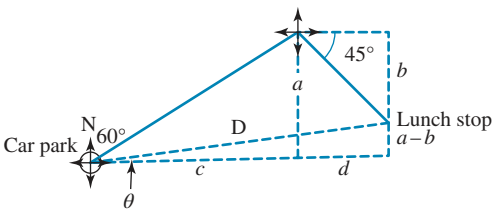
   c.



9. a. 13.38 km
   b. 14.86 km
   c. 222°T
   d.



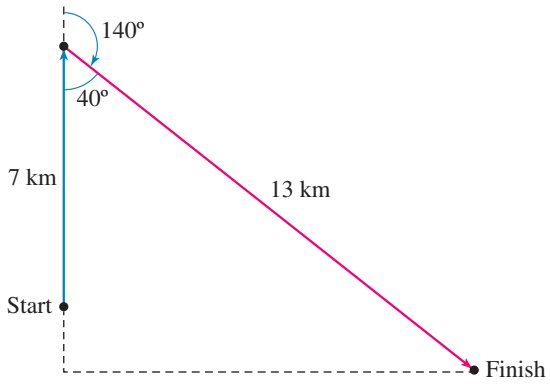
   e. 51.42 km
   f. 61.28 km
   g. 310°T
10. 215°T
11. 1.732 km
12. a. 9.135 km b. 2.305 km c. 104°10′T
13. 684.86 km
14. 6.10 km and 239°T
15. 111°T
16. a. $(180 + \theta)$ °T b. $(\theta - 180)$ °T
17. a. 27.42 km b. N43°W or 317°T
18. a.



   b. 1.76 km North
   c. 14.63 km East
   d. N83.15°E
   e. D = 14.74 km
19. 3.65 km on a bearing of 108°T

5.9 Applications

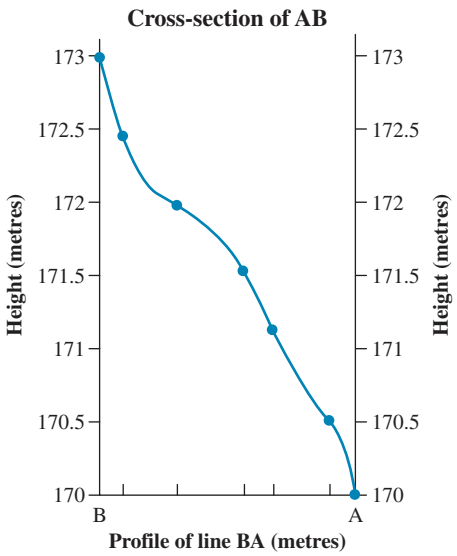
1. 6.09 m
2. 62°33′
3. a. 1.82 m b. 27.78 m
4. 31°49′
5. 5.94 m
6. 49 m
7. 278.49 cm
8. 288.68 m
9. 194 m
10. 1.829 km
11. a. 11°32′ b. 4°25′
12. a. 35.36 cm b. 51.48 cm c. 51.48 cm
    d. 57.23 cm e. 29°3′ f. 25°54′
    g. 25.74 cm h. 12.5 cm i. 25°54′
    j. 28.61 cm
13. a. 77° b. 71°56′ c. 27.35 cm
14. a. 7.05 cm b. 60°15′ c. 8.12 cm
15. a. 28.74 cm b. 40.64 cm c. 66°37′
16. $\sin(\theta) = \frac{O}{H}$. Since the hypotenuse H is the longest side in the right-angled triangle, when dividing O by H the value will be between 0 and 1.
17. a. 122.97° b. 142.37°
18. a. 90 m
    b. $h = \frac{d\tan(\theta_1)}{\tan(\theta_1) + \tan(\theta_2)} \times \tan(\theta_2)$
    c. 250 m
19. a.



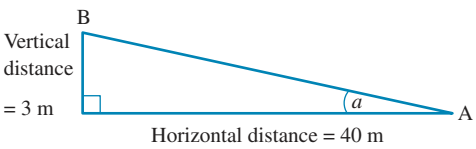
    b. 2.96 km
    c. 110°
20. The golfer must hit the ball 324.4 m at an angle of 9.7° off the direct line.
21. a. 7.69 cm b. 6.91 cm

Project

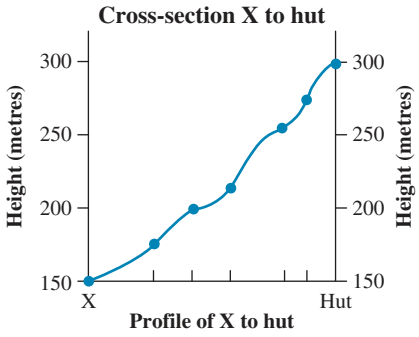
1.



2. a. 8 cm b. 40 m
3. 3 m
4.



5. $a = 4°17'$
6. Gradual to moderate
7.



The average slope is 11.46° — moderate to steep.

5.10 Review questions

1. D
2. D
3. D
4. D
5. D
6. B
7. D
8. B
9. B
10. A

11. **a.** $x = 113.06$ cm **b.** $x = 83.46$ mm

12. 9.48 cm

13. 8.25 mm

14. 17.6 m

15. 26.86 m

16. 67.98 km

17. 4.16 km

18. $40°32'$

19. **a.** 11.04 cm **b.** 15.61 cm **c.** $59°1'$

20. **a.** 1280.64 m

b. 2 minutes, 16.3 seconds past 12 pm

21. 33.29 m, 21.27 m

22. **a.** $4\sqrt{2}b$

b. 19.5°, 70.5°, 90°.

c. $\text{Area} = \frac{1}{2} \text{base} \times \text{height}$

$= \frac{1}{2} \times 2b \times 4\sqrt{2}b$

$= 4\sqrt{2}b^2 \text{ cm}^2$

23. **a.** 60° **b.** 3.98 km **c.** 71.5° **d.** 1.33 km